

# **QUANTUM FIELD THEORY OF TIME**

The Weak Entanglement Symmetry Hypothesis

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# Abstract

This work addresses the frozen-time problem of canonical quantum gravity: the Wheeler–DeWitt constraint eliminates evolution from the fundamental description, while standard quantum theory treats time as an external parameter. To resolve this tension, we propose the Weak Entanglement Symmetry Hypothesis (WESH), which promotes entanglement as the carrier of statistical Lorentz-type symmetry and as an engine of endogenous dissipative dynamics. Such symmetry is recovered in expectation over the Poissonian ensemble of *eigentime* events: discrete stochastic localizations of the time-field operator  $\hat{T}(x)$  within its continuous spectrum, induced by the quadratic WESH dissipators  $\mathcal{D}[\hat{T}^2(x)]$  and  $\mathcal{D}[L_{xy}]$ . Global conserved charges obey a stronger operator-level constraint (WESH–Noether,  $\mathcal{L}^\dagger[\hat{Q}_a] = 0$ ).

On this foundation, physical time becomes a local quantum field operator  $\hat{T}(x)$ , subject to superposition and objective collapse, with dynamics constructed from first principles. An endogenously dissipative master equation in an auxiliary, non-observable label  $s$  generates physical time through  $dt/ds = \Gamma[\rho] \geq 0$ . Complete positivity, finite-range locality, CPT symmetry, and a pre-geometric WESH–Noether conservation principle single out a CPT-even GKSL structure with quadratic local dissipators  $\mathcal{D}[\hat{T}^2(x)]$  and a Rényi-2 weighted bilocal channel. The dissipative kernel  $\gamma(x, y)$  is defined via a symmetric, pre-geometric finite-range proximity relation  $N_\xi$  of range  $\xi \simeq L_P$ , without background metric input.

The framework has no free dimensionless parameters and yields two quantitative signatures: collective stability scaling  $\tau_{\text{coh}} \propto N^2$  and a robust  $\cos^2 \theta$  angular law for parity decay. To test these predictions, we conducted both CPU-based collision-model simulations ( $N = 2\text{--}16$ ) and large-scale experiments on IBM Eagle and Rigetti Ankaa-3 NISQ processors. Both lines of evidence show trends consistent with the predicted signatures.

# 1 QFTT Framework & WESH

## 1.1 Problem of time and quantum gravity strategy

Most approaches to quantum gravity (loop quantum gravity, Ashtekar et al., 1995; string theory; causal dynamical triangulations) either quantize an already geometric spacetime or work semiclassically on a fixed background, leaving the operational status of time itself unresolved. Alternative proposals such as the Page–Wootters mechanism (Page & Wootters, 1983) tie the emergence of time to entanglement between system and clock but do not derive dissipative dynamics or metric structure.

Here we address both issues at once: we promote physical time to a local quantum field operator  $\hat{T}(x)$  on an extended kinematical Hilbert space, evolve the state by a completely positive WESH master equation in an auxiliary ordering parameter  $s$ , and let physical time emerge from eigentime events and entanglement-weighted dissipative flow.

## 1.2 Time field and canonical structure

The elevation of physical time to a field operator follows from the canonical structure of constrained systems.

In the minisuperspace approximation, the Wheeler–DeWitt constraint decomposes as

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{universe}} + \hat{P}_T \approx 0, \quad [\hat{T}, \hat{P}_T] = i\hbar, \quad (1.1)$$

where  $\hat{P}_T$  is the momentum conjugate to a global time variable. The constraint does not forbid dynamics; it enforces energy balance between the time sector and the remaining degrees of freedom. Passing to the local field-theoretic extension, we postulate canonical commutation relations on an arbitrary kinematical slice  $\Sigma$  (since the algebra is independent of the choice of coordinates prior to metric emergence):

$$[\hat{T}(\mathbf{x}), \hat{\Pi}_T(\mathbf{y})] = i\hbar \delta_{\Sigma}^{(3)}(\mathbf{x} - \mathbf{y}), \quad [\hat{T}, \hat{T}] = [\hat{\Pi}_T, \hat{\Pi}_T] = 0. \quad (1.2)$$

On the constraint surface, the conjugate momentum is identified with (minus) the local energy density,

$$\hat{\Pi}_T(\mathbf{x}) = -\hat{\mathcal{H}}(\mathbf{x}), \quad [\hat{\mathcal{H}}(\mathbf{x}), \hat{T}(\mathbf{y})] = i\hbar \delta_{\Sigma}^{(3)}(\mathbf{x} - \mathbf{y}), \quad (1.3)$$

so that the local Hamiltonian constraint  $\hat{\mathcal{H}}(x) + \hat{\Pi}_T(x) \approx 0$  becomes the generator of local time translations. This identification transforms the Wheeler–DeWitt

“frozen time” into a dynamical equilibrium condition: the constraint balances the quantum time frame against the local energy content.

The spectral decomposition

$$\hat{T}(x) |t\rangle_x = t |t\rangle_x, \quad \int_{-\infty}^{+\infty} dt |t\rangle_x \langle t| = \mathbb{1}_x \quad (1.4)$$

allows one to expand the global state as a superposition over temporal configurations:

$$|\Psi\rangle = \int \mathcal{D}t(x) \mathcal{D}\phi \Psi[t(x), \phi] \bigotimes_x |t(x)\rangle_x \otimes |\phi\rangle. \quad (1.5)$$

In the WDW sector, temporal indeterminacy is therefore physical, not artifactual: the universe exists in a quantum superposition of chronologies. Resolving this superposition requires a dynamical mechanism that cannot be unitary in the conventional sense, there is no external  $t$  to generate a Schrödinger flow, but must be dissipative. The structure of this dissipation is fixed by symmetry, as we now explain.

### 1.3 The Weak Entanglement Symmetry Hypothesis

In this framework “weak” does not refer to a small coupling, but to the level at which Lorentz-type symmetry is enforced. Lorentz invariance is not required pointwise on each individual collapse trajectory, but is recovered in expectation over the Poissonian ensemble of eigentime events. Global conserved charges obey a stronger, operator-level constraint (Sec. 1.4). Entanglement correlations  $C(\Psi; x, y)$  distribute the effect of a local collapse across all correlated regions, so that apparent local violations are globally compensated when averaged over the process. Such symmetry is therefore a property of the process measure rather than of individual realizations.

### 1.4 WESH–Noether conservation

In standard field theory, Noether’s theorem guarantees that every continuous symmetry of the action corresponds to a conserved quantity. The WESH framework operates under non-unitary, dissipative dynamics where the classical action principle does not apply directly. Nevertheless, the entanglement structure of the WESH generator supports an analogous conservation law: a multi-local symmetry of the entangled state (a transformation applied simultaneously to multiple subsystems

that leaves the joint state invariant) implies the existence of a conserved charge whose expectation value remains constant along the auxiliary flow. This conservation holds in expectation over the stochastic ensemble, so that while individual eigentime events may redistribute charge locally, the system as a whole preserves it exactly. We call this principle *WESH–Noether conservation*.

In the pre-geometric regime there is no background metric: the constraint  $\hat{H}_{\text{tot}}|\Psi\rangle = 0$  removes the generator of time translations, leaving only algebraic structure. In a closed universe there is no external bath to absorb conserved charges, so the generator must preserve them exactly:

$$\mathcal{L}^\dagger[\hat{Q}_a] = 0 \quad \forall a. \quad (1.6)$$

This *WESH–Noether conservation* is pre-geometric: it is formulated in the auxiliary parameter  $s$ , before physical time or spacetime have emerged, and guarantees path independence in state space, so that  $\langle \hat{Q}_a \rangle$  does not depend on the particular trajectory  $\rho(s)$ .

**Algebraic mechanism.** The conservation law (1.6) translates into commutant conditions on the generator. For Hermitian Lindblad operators  $L$ , the adjoint dissipator acts on observables as

$$\mathcal{D}_L^\dagger[\hat{Q}] = -\frac{1}{2}[L, [\hat{Q}, L]]. \quad (1.7)$$

Under the WESH structure ( $\nu > 0$ ,  $\gamma(x, y) C(\Psi; x, y) \geq 0$ , Hermitian jumps) and mild spectral regularity (Appendix D, Prop. D.2),  $\mathcal{L}^\dagger[\hat{Q}_a] = 0$  is equivalent to

$$\begin{aligned} [H_{\text{eff}}, \hat{Q}_a] &= 0, & [\hat{T}^2(x), \hat{Q}_a] &= 0, \\ [L_{xy}, \hat{Q}_a] &= 0 \quad \text{for all } (x, y) \text{ with nonzero rate.} \end{aligned} \quad (1.8)$$

The double commutator (1.7) is the algebraic engine of conservation: it vanishes precisely when each Lindblad channel commutes with the charge.

**Difference structure.** For the bilocal channel, the commutant condition together with the requirement of pairwise-balanced redistribution (zero net injection) selects the difference form,

$$L_{xy} = \hat{T}^2(x) - \hat{T}^2(y). \quad (1.9)$$

This structure redistributes coherence between  $N_\xi$ -connected points without injecting or removing total charge: any increase at  $x$  is compensated by a decrease at  $y$ . The local channel  $\mathcal{D}[\hat{T}^2(x)]$  satisfies the same commutant provided the time field is *T-neutral*,

$$[\hat{T}(x), \hat{Q}_{\text{tot}}] = 0 \quad \forall x. \quad (1.10)$$

T-neutrality is not imposed externally; it is the unique condition compatible with path independence, universal time emergence, and no-signaling (Appendix D, Remark D.4).

**Theorem 1.1** (WESH–Noether). *Let  $\mathcal{L}$  be a GKSL generator with Hermitian jump operators  $L_\alpha \in \{\hat{T}^2(x), L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)\}$  and let  $\hat{Q}_a$  be any global charge. Then the following are equivalent:*

- (i)  $\mathcal{L}^\dagger[\hat{Q}_a] = 0$  (generator-level conservation);
- (ii)  $[H_{\text{eff}}, \hat{Q}_a] = 0$  and  $[L_\alpha, \hat{Q}_a] = 0$  for all  $\alpha$  (commutant conditions);
- (iii) for every trajectory  $\rho(s)$  one has

$$\frac{d}{ds} \langle \hat{Q}_a \rangle_{\rho(s)} = \text{Tr}(\mathcal{L}^\dagger[\hat{Q}_a] \rho(s)) = 0, \quad (1.11)$$

(path independence in  $s$ ).

Under T-neutrality (1.10), all three conditions hold for every conserved charge.

This theorem extends Noether’s result to dissipative quantum dynamics: constants of motion are identified with fixed points of the Heisenberg semigroup generated by  $\mathcal{L}^\dagger$ . The proof uses the double-commutator identity (1.7) together with mild spectral regularity assumptions.

**Functional role in the present framework.** In the remainder of this section, WESH–Noether is used as a selection principle: together with CPT symmetry and collective stability, it fixes the admissible form of the dissipative generator and singles out the quadratic WESH structure employed in Eq. (1.15). The same constraint plays a second role in the infrared analysis. There, its pre-geometric character becomes essential: the WESH flow admits a unique fixed point that minimizes a Lyapunov functional subject to the Noether constraints, and at this fixed point the gradients of the time field and of the entanglement potential align,  $\partial_\mu \tau = k \partial_\mu \Phi$  (with  $\tau := \langle \hat{T} \rangle_\rho$ ). The emergent Einstein–Hilbert dynamics and hidden-sector cancellation follow from this aligned fixed point at which gradient alignment yields emergent geometry. Both applications derive from the same algebraic condition; only the physical interpretation shifts from constraint selection to geometric emergence.

The complete operator-level derivation, including spectral regularity conditions and the bridge to alignment, is developed in Appendix D.

## 1.5 From Wheeler–DeWitt to WESH: constraint closure

The question of which dissipative completion is admissible in the Wheeler–DeWitt sector arises naturally at this stage. In a closed universe no external bath can absorb conserved charges, no external clock can parametrize collapse, and no background metric can define locality in advance. Any non-unitary extension must therefore arise internally, as an effective redistribution of information among relational degrees of freedom within the WDW kinematics, where no net charge or energy can be injected. We seek the most general GKSL-type extension compatible with the requirements collected in Table 1.

| Requirement                                                                   | Excludes                                                          | Selects                                                                                                  |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| CPT-evenness +<br>collective stability<br>( $\tau_{\text{coh}} \propto N^2$ ) | odd local jumps;<br>higher even monomials ( $n > 1$ )             | quadratic local channel<br>$\mathcal{D}[\hat{T}^2(x)]$                                                   |
| WESH +<br>closed-universe<br>conservation                                     | entanglement-blind channels;<br>unbalanced bilocal redistribution | bilocal difference form<br>$L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)$ ;<br>entanglement gate $C(\Psi; x, y)$ |
| Pre-geometric locality<br>+ bounded collapse<br>power                         | metric-dependent support;<br>unbounded total rate                 | finite-range kernel $K_\xi \not\propto_{N_\xi}$ ;<br>normalization $\gamma \propto N^{-2}$               |

**Table 1:** Constraint ledger for the WESH generator. Three groups of requirements converge to select a single admissible structural class.

The derivation is eliminative: each constraint removes a class of alternatives, and their intersection leaves a single admissible generator. We now trace this closure step by step.

**Local channel: CPT symmetry and collective stability.** Under time reversal  $\hat{T} \rightarrow -\hat{T}$ , any admissible local Lindblad operator must be an even function of the field:  $L_x = F(\hat{T}(x))$  with  $F(z) = \sum_{m \geq 1} a_{2m} z^{2m}$ . This CPT-evenness excludes all odd-power collapse channels, including the linear operator  $\hat{T}(x)$  itself. Within the surviving even class, collective stability sharpens parity into uniqueness. Anticipating the pairwise normalization derived below, under  $\gamma \propto N^{-2}$  a monomial  $F(\hat{T}) \sim \hat{T}^{2n}$  induces coherence scaling  $\tau_{\text{coh}}(N) \propto N^{2n}$ :

$$\text{If } L_x \sim \hat{T}^{2n} \quad \Rightarrow \quad \tau_{\text{coh}}(N) \propto N^{2n}. \quad (1.12)$$

The structural requirement  $\tau_{\text{coh}}(N) \propto N^2$  therefore fixes  $n = 1$ . The admissible local normal form is  $\mathcal{D}[\hat{T}^2(x)]$ . Independently, a Wilsonian IR argument reaches the same conclusion: among CPT-even local monomials, the lowest-dimension term dominates at long wavelengths, while higher even powers are irrelevant deformations. The convergence of two independent selection mechanisms, structural (bootstrap closure) and effective (IR dominance), strengthens confidence in the uniqueness of the quadratic channel.

Beyond scaling, the selection  $n = 1$  rests on a deeper algebraic argument that we condense here; its full development belongs to a forthcoming paper on Quantum Self-Interacting Poisson Structures (QSIPS).

**Remark 1.1** (Algebraic root of quadratic uniqueness). *For any  $n \geq 1$ , the bilocal jump operator factorizes as*

$$\hat{T}^{2n}(x) - \hat{T}^{2n}(y) = [\hat{T}^2(x) - \hat{T}^2(y)] \cdot P_{n-1}(\hat{T}^2(x), \hat{T}^2(y)),$$

where  $P_{n-1}$  is the symmetric polynomial  $P_0 = 1$ ,  $P_1 = \hat{T}^2(x) + \hat{T}^2(y)$ , etc. (the identity is elementary and holds for any commuting self-adjoint operators). For  $n = 1$  the jump is irreducible: pure mismatch with no field-dependent prefactor. For  $n \geq 2$  it is reducible, and the polynomial  $P_{n-1}$  contaminates every structure built on  $L_{xy}$ . Setting  $u(x) := \langle \hat{T}^2(x) \rangle_{\rho^*}$  for the fixed-point expectation:

- (a) Lyapunov obstruction. *The bilocal part of the WESH Lyapunov functional  $\mathcal{M}$  becomes a weighted energy with effective stiffness  $\propto u^{2(n-1)}$ , which degenerates where  $u \rightarrow 0$ ; the Dirichlet form (constant stiffness, strict convexity, unique minimizer) is lost.*
- (b) Field equation obstruction. *The infrared Euler–Lagrange equation becomes quasilinear (field-dependent principal part) rather than semilinear; degenerate ellipticity at  $u \rightarrow 0$  spoils uniqueness and regularity of stationary solutions.*
- (c) Hidden-sector obstruction. *The residual stress tensor after cancellation at the fixed point carries a coefficient  $(4n-2) \delta\tau/\tau_*$ , growing with  $n$  and degrading the universality of the emergent Einstein equation.*

*All three obstructions trace to the single algebraic fact of jump reducibility. The quadratic selection is therefore load-bearing at every level of the construction.*

**Bilocal sector: WESH and closed-universe conservation.** A purely local dissipator  $\mathcal{D}[\hat{T}^2(x)]$  is entanglement-blind: it applies the same rate regardless of the



correlation structure between sites. WESH posits that entanglement is the engine of dissipation, so a channel that does not see correlations cannot realize the hypothesis. The dissipative extension must therefore include a bilocal sector coupling correlated pairs. In a closed universe, WESH–Noether conservation ( $\mathcal{L}^\dagger[\hat{Q}_a] = 0$ ) then fixes the form of this sector: the commutant conditions (1.8) require a balanced redistribution, selecting the difference form

$$L_{xy} = \hat{T}^2(x) - \hat{T}^2(y), \quad (1.13)$$

where any gain at  $x$  is compensated by a loss at  $y$ . The bilocal channel must further be gated by the entanglement indicator  $C(\Psi; x, y)$  (Eq. (1.21)), which vanishes on product reductions  $\rho_{xy} = \rho_x \otimes \rho_y$  and activates only on genuinely correlated pairs, ensuring that factorized sectors remain dynamically decoupled.

**Finite-range pre-geometric locality and bounded collapse power.** Since no background metric is available at the fundamental level, admissibility cannot be defined by light-cone data. The kernel  $\gamma(x, y)$  must instead be supported on the symmetric pre-geometric proximity relation  $N_\xi$  (Definition 1.1), with a nonnegative short-range envelope  $K_\xi$  that is positive and  $L^1$ -integrable. This fixes the locality class of the generator without importing causal ordering. With  $\mathcal{O}(N^2)$  active pairs, the bilocal rate must then scale as  $N^{-2}$  to keep the total collapse power bounded while preserving a non-vanishing macroscopic effect.

**Theorem 1.2** (Constraint closure of the WESH generator). *Consider GKSL-type dissipative completions of the Wheeler–DeWitt sector in an auxiliary ordering parameter  $s$ . Impose CPT-evenness, collective stability  $\tau_{\text{coh}}(N) \propto N^2$ , entanglement-mediated dissipation (WESH) with closed-universe charge conservation (WESH–Noether), product-sector consistency, and finite-range pre-geometric locality with bounded total collapse power. Then the admissible dissipative sector is uniquely of WESH type: a local quadratic channel  $\mathcal{D}[\hat{T}^2(x)]$ , a bilocal difference channel  $\mathcal{D}[L_{xy}]$  with  $L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)$ , a nonnegative finite-range kernel  $\gamma(x, y) = \frac{\gamma_0}{N^2} K_\xi(x-y) \mathbb{1}_{N_\xi}(x, y)$ , and a state-dependent entanglement gate  $C(\Psi; x, y)$ .*

*Proof sketch.* CPT excludes odd local jumps; collective stability fixes the quadratic normal form ( $n = 1$ , Eq. (1.12)). WESH excludes entanglement-blind completions, forcing a bilocal sector; WESH–Noether fixes its balanced form (Sec. 1.4). Product-sector consistency forces state-dependent activation. Pre-geometric locality restricts the kernel to  $N_\xi$  (Definition 1.1); bounded collapse power fixes  $\gamma \propto N^{-2}$ . No larger class survives all constraints simultaneously.  $\square$

The constraint analysis therefore does not merely motivate the WESH structure; it fixes the admissible generator class. We now write the resulting dissipative completion of the Wheeler–DeWitt sector explicitly.

## 1.6 The WESH master equation

The constraints of Theorem 1.2 determine the dissipative sector of the generator. Before writing it explicitly, we fix the integration conventions.

**Measure conventions.** We use coarse-grained integration measures normalized to the correlation four-volume:

$$\int_x := \frac{1}{V_\xi} \int d^4x, \quad \int_{xy} := \frac{1}{V_\xi^2} \int d^4x d^4y, \quad V_\xi = \mathcal{O}(\xi^4). \quad (1.14)$$

With these conventions,  $\nu$  carries dimensions of inverse time; Prop. 1.1 uses unnormalized integrals, so  $\nu$  and  $\int d^4y \gamma(x, y)$  scale as  $\mathcal{O}(\gamma_0 V_\xi)$  and  $\mathcal{O}(\gamma_0 V_\xi / N^2)$ , respectively.

**Generator.** The GKSL generator compatible with these constraints takes the form

$$\begin{aligned} \frac{d\rho}{ds} = & -i[\hat{H}_{\text{eff}}, \rho] + \nu \int \frac{d^4x}{V_\xi} \mathcal{D}[\tilde{T}^2(x)] \rho \\ & + \iint \frac{d^4x d^4y}{V_\xi^2} \gamma(x, y) C(\Psi; x, y) \mathcal{D}[L_{xy}] \rho \end{aligned} \quad (1.15)$$

where  $\tilde{T} = \hat{T}/\tau_s$  is the dimensionless field and  $V_\xi \sim \xi^4$  sets the coarse-graining four-volume. The effective Hamiltonian splits as

$$\hat{H}_{\text{eff}} = \hat{H}_{\text{local}} + \hat{H}_{\text{ent}}, \quad (1.16)$$

with  $\hat{H}_{\text{local}}$  collecting matter and time-field contributions and  $\hat{H}_{\text{ent}}$  the entanglement-mediated correction.

The dissipative part uses the standard Lindblad superoperator

$$\mathcal{D}[O] \rho \equiv O \rho O^\dagger - \frac{1}{2} \{O^\dagger O, \rho\}, \quad (1.17)$$

so that the local channel is generated by the Hermitian jump operator  $\tilde{T}^2(x)$ ,

$$\mathcal{D}[\tilde{T}^2(x)] \rho = \tilde{T}^2(x) \rho \tilde{T}^2(x) - \frac{1}{2} \{\tilde{T}^4(x), \rho\}, \quad (1.18)$$

and can be interpreted as the continuous counterpart of the conditional update

$$\rho \longrightarrow \frac{\hat{T}^2(x) \rho \hat{T}^2(x)}{\text{Tr}[\hat{T}^2(x) \rho \hat{T}^2(x)]} \quad (1.19)$$

at the point  $x$ .

### Finite-range dissipative kernel.

**Definition 1.1** (Pre-geometric neighbourhood). *Let  $N_\xi(x)$  denote a finite-range neighbourhood on the label-space:  $y \in N_\xi(x)$  whenever the pair  $(x, y)$  satisfies a pre-geometric proximity relation with range  $\xi$ . The relation is symmetric ( $y \in N_\xi(x) \Leftrightarrow x \in N_\xi(y)$ ) and carries no causal or temporal ordering. No background metric is required; the relation may be topological, combinatorial, or defined on a graph.*

The bilocal weight combines exponential falloff, finite-range admissibility, and collective suppression:

$$\gamma(x, y) = \frac{\gamma_0}{N^2} K_\xi(x - y) \mathbb{K}_{N_\xi}(x, y), \quad (1.20)$$

with  $K_\xi(z)$  a short-range profile of range  $\xi \simeq L_P$  (e.g.  $K_\xi(z) = e^{-d(z)/\xi}$ ),  $\gamma_0 > 0$  a rate scale, and  $\mathbb{K}_{N_\xi}$  the indicator of the pre-geometric neighbourhood. The  $N^{-2}$  prefactor ensures that the sum over all pairs yields a total rate of order  $\gamma_0$ , independent of system size. In the continuum regime where the emergent metric  $g_{\mu\nu}$  is available,  $N_\xi(x)$  coincides with the causal domain of  $x$  up to corrections  $\mathcal{O}(\xi/L)$ ; the pre-geometric definition requires only finite-range locality. Three distinct objects enter the construction: the rate envelope  $K_\xi$  (positive,  $L^1$ , finite-range), which weights the dissipative channel; the Yukawa mediator  $G_\xi$ , which defines the entanglement potential  $\Phi$  (Appendix A); and the admissibility projector  $\mathbb{K}_{N_\xi}$ , which encodes locality without metric input.

**Terminological note.** Two kernels appear with distinct roles. The function  $\gamma(x, y)$  in Eq. (1.20) is the exponential finite-range dissipative weight entering the generator (1.15). A Yukawa-type kernel  $G_\xi(x - y)$ , introduced in Appendix A, defines an entanglement potential  $\Phi(x) = \int d^4y G_\xi(x - y) C(\Psi; x, y)$  and enters

the metric-emergence analysis. Both kernels have the same short range  $\xi \simeq L_P$ , but  $\gamma$  weights collapse rates whereas  $G_\xi$  averages correlations for the emergent geometry.

**Entanglement gate.** The bilocal channel is modulated by a Rényi-2 entanglement indicator,

$$C(\Psi; x, y) = [\text{Tr } \rho_{xy}^2 - \text{Tr } \rho_x^2 \text{Tr } \rho_y^2]_+ \in [0, 1), \quad [u]_+ := \max(u, 0). \quad (1.21)$$

For comparison with standard connected correlators, one may also define

$$C_{\text{loc}}(\Psi; x, y) = \langle \hat{f}(x) \hat{f}(y) \rangle_\Psi - \langle \hat{f}(x) \rangle_\Psi \langle \hat{f}(y) \rangle_\Psi, \quad (1.22)$$

for a local observable  $\hat{f}(x)$ . This quantity is used only as a diagnostic; it never appears in the weights of Eq. (1.15). The  $[\cdot]_+$  construction ensures  $0 \leq C < 1$  in all cases:  $C = 0$  on product states automatically, and  $C \rightarrow 1$  as  $\rho_{xy}$  approaches a pure entangled state. The gate is Lipschitz in the trace norm of the reduced states, since  $\text{Tr } \rho^2$  is polynomial in  $\rho$  (Lipschitz on bounded sets), products of bounded Lipschitz functions are Lipschitz, and  $[\cdot]_+$  is 1-Lipschitz. When  $x$  and  $y$  are maximally entangled,  $C \rightarrow 1$  and the synchronization channel operates at full strength; for product states,  $C = 0$  and the points evolve independently. This state-dependent weighting is the core of the Weak Entanglement Symmetry Hypothesis: geometry emerges from the entanglement structure, not the reverse.

**Difference form.** The bilocal Lindblad operator

$$L_{xy} = \tilde{T}^2(x) - \tilde{T}^2(y) \quad (1.23)$$

is simply the rescaled version of the difference form  $L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)$  of Eq. (1.9), using the dimensionless field  $\tilde{T}(x) = \hat{T}(x)/\tau_s$ . It penalizes temporal desynchronization between  $N_\xi$ -connected points. Its difference structure guarantees that no net charge is introduced or removed by the bilocal channel, a property that will later underpin the pre-geometric conservation principle.

**No-signaling.** Because  $\gamma(x, y)$  vanishes for pairs outside  $N_\xi$ , the bilocal generator has  $N_\xi$ -local support. For any region  $A \subset \Sigma$  with complement  $A^c$ , the reduced state obeys

$$\frac{d}{ds} \rho_A(s) = \text{Tr}_{A^c} (\mathcal{L}_{N_\xi(A)}[\rho(s)]), \quad (1.24)$$

where  $\mathcal{L}_{N_\xi(A)}$  denotes the restriction of the generator to operators supported in the  $N_\xi$ -neighbourhood of  $A$ . Hence operations on  $A^c$  outside  $N_\xi$  cannot instantaneously affect  $\rho_A$  (Appendix E).

Finite-range locality is thus built into the generator itself at the pre-geometric level:  $N_\xi$ -locality is a property of the WESH construction, not of the emergent metric. At the alignment fixed point,  $N_\xi$ -separation coincides with metric space-like separation (Remark 1.3), and the pre-geometric no-signaling recovers standard relativistic no-signaling on Cauchy slices (Appendix E, Proposition E.1).

**Remark 1.2** (Nonlinearity, CP/TP, and no-signaling). *Under the structural assumptions of the WESH generator (Hermitian jumps, nonnegative rates, finite-range kernel), freezing the state-dependent gate  $C(\Psi; x, y)$  on each micro-interval yields a standard GKSL generator with Hermitian jumps and nonnegative rates, so that each step of the WESH evolution is completely positive, trace preserving, and  $N_\xi$ -local (McCauley et al., 2020). The state dependence of  $C$  can then be viewed as inducing a time-dependent generator; complete positivity and trace preservation for the resulting non-autonomous dynamics follow from product-integral constructions for time-dependent GKSL evolutions (Rivas & Huelga, 2012). Appendix E provides the full operator-level proof of CP/TP and no-signaling for Eq. (1.15).*

## 1.7 Bootstrap, Eigentimes, and the Emergence of Spacetime

In this subsection we analyse how a directed temporal ordering, and subsequently spacetime geometry, may emerge once time is promoted to a local quantum field operator  $\hat{T}(x)$  in the Wheeler–DeWitt sector. Spatial geometry, though not quantized directly, emerges from the alignment dynamics governed by entanglement correlations and the finite-range kernel. The construction must generate temporal order and geometric structure without reintroducing an external time parameter or any pre-assigned temporal layering.

Throughout this subsection  $\langle \cdot \rangle_{\rho(s)}$  denotes expectation in the state  $\rho(s)$ ,  $\Delta\hat{T}(x) := \hat{T}(x) - \langle \hat{T}(x) \rangle_{\rho(s)}$ , and we set  $\hbar = c = 1$ .

**Architectural necessity.** Once time is encoded by the operator  $\hat{T}(x)$ , it cannot simultaneously act as an external evolution parameter; introducing a separate clock would restart the hyper-time regress. Any emergent temporal order must therefore arise *endogenously*, from the statistics and structure of collapse events of  $\hat{T}(x)$  itself.

A theory that aims to generate spacetime from a setting that, by necessity, lies *prior to* geometric structure, and simultaneously excludes any external evolution parameter, must satisfy strict architectural constraints tied to the endogenous

handling of information. We may therefore identify the following prohibitions:

- *Sequential* constructions presuppose a pre-existing temporal ordering;
- *Hierarchical* constructions presuppose layered or stratified temporal ordering;
- *Iterative* algorithms presuppose discrete temporal steps.

The bootstrap is not a modeling choice but the *only self-sufficient mathematical architecture* compatible with these constraints. This places our construction in continuity with earlier self-consistency programmes: Chew’s *S*-matrix bootstrap (Chew, 1961) and the conformal bootstrap of Rattazzi, Rychkov, Tonni, and Vichi (2008), where physical quantities are fixed by global consistency rather than external inputs. Here, the same philosophy governs *chronogenesis* and the induced emergence of spacetime structure. In short, the bootstrap provides a self-consistent, atemporal configuration in which the Markovian dynamics and Poissonian stochasticity conspire to generate temporal order and geometric structure without any external temporal input or historical memory.

**Bootstrap as an atemporal configuration.** In WESH, the bootstrap is *not* a temporal sequence. It is an atemporal, non-hierarchical configuration in which multiple elements co-determine one another: a simultaneous fixed-point condition rather than a deterministic chain. To illustrate (as *mutual constraints*, not chronological stages): the quantum state fixes entanglement correlations  $C(\Psi; x, y)$ ; these correlations weight dissipative channels; stochastic eigentime events realize localized outcomes of the time field; the concentration of such events constitutes the emergent spacetime arena in which the state is defined. No element is “first”; each requires the others for consistency.

This dissolves apparent circularities. As in the Page–Wootters mechanism (Page & Wootters, 1983), where a timeless global state encodes relational temporal correlations, and as in Wheeler–DeWitt, which admits no “before” once time emerges, here there is no “before the first eigentime” because eigentimes *constitute* the temporal structure. Questions that presuppose a pre-existing sequence are simply ill-posed. Combined with truly stochastic localization (which carries no temporal memory) and with the finite-range kernel  $\gamma(x, y)$  enforcing a finite memory window  $\tau_{\text{corr}} \ll \tau_{\text{Eig}}$ , this yields the *minimal sufficient* pre-geometric framework: the bootstrap eliminates external temporal scaffolding; Poissonian statistics eliminate dynamical memory; the short-range kernel coordinates the process without long-term history dependence.

**Eigentimes as time quanta and geometric seeds.** The basic quanta of this construction are *eigentimes*: stochastic localizations of the time field at definite outcomes of the spectral measure of  $\hat{T}(x)$ . Let  $E_x(dt)$  be the spectral measure of

$\hat{T}(x)$ , so that

$$\hat{T}(x) = \int_{\mathbb{R}} t E_x(dt), \quad \mathbb{K}_x = \int_{\mathbb{R}} E_x(dt). \quad (1.25)$$

For a local time-sector state  $|\psi_T(x)\rangle$ , the spectral expansion reads

$$|\psi_T(x)\rangle = \int_{\mathbb{R}} E_x(dt) |\psi_T(x)\rangle, \quad (1.26)$$

and an eigentime event in a Borel window  $\Delta \subset \mathbb{R}$  is modeled operationally by the localization

$$|\psi_T(x)\rangle \longrightarrow \frac{E_x(\Delta) |\psi_T(x)\rangle}{\sqrt{\langle \psi_T(x) | E_x(\Delta) | \psi_T(x) \rangle}}, \quad p_x(\Delta) = \langle \psi_T(x) | E_x(\Delta) | \psi_T(x) \rangle. \quad (1.27)$$

Eigentimes are discrete stochastic realizations of the continuous spectrum of  $\hat{T}(x)$ , not quanta of space. Spatial geometry is induced indirectly: the pattern of eigentime events, weighted by entanglement via  $C(\Psi; x, y)$  and correlated by the finite-range kernel  $\gamma(x, y)$ , seeds the smooth spacetime geometry reconstructed at the dynamical fixed point of the WESH flow. Regions where eigentimes concentrate densely support a well-defined geometric structure; sparsity leaves geometry weakly defined.

The mean production rate of eigentimes along the auxiliary flow is

$$\begin{aligned} \left\langle \frac{dN_{\text{Eig}}}{ds} \right\rangle &= \frac{\Gamma[\Psi(s)]}{\tau_{\text{Eig}}} \\ &= \nu \int \frac{d^4x}{V_\xi} \text{Tr} [\tilde{T}^2(x) \rho(s) \tilde{T}^2(x)] \\ &\quad + \iint \frac{d^4x d^4y}{V_\xi^2} \gamma(x, y) C(\Psi; x, y) \text{Tr} [L_{xy}^\dagger L_{xy} \rho(s)], \end{aligned} \quad (1.28)$$

with  $\nu > 0$  a state-independent intensity scale. The local hazard (the instantaneous propensity at point  $x$ ) satisfies

$$\lambda(x) = \nu \text{Tr} [\tilde{T}^2(x) \rho \tilde{T}^2(x)] \geq \nu (\text{Var}_\rho[\tilde{T}(x)])^2, \quad (1.29)$$

(Cauchy–Schwarz). Since  $\tilde{T}(x) = \hat{T}(x)/\tau_s$  shares the same spectral projectors as  $\hat{T}(x)$ , the activation condition can equivalently be expressed in terms of the physical variance of  $\hat{T}(x)$ :

$$\langle (\Delta \hat{T}(x_0))^2 \rangle_{\rho(s)} > \tau_{\text{thr}}^2 \implies \text{eigentime at } x_0. \quad (1.30)$$

Two scales control the statistics:

$$\tau_{\text{Eig}} \equiv \nu^{-1}, \quad \tau_{\text{ref}}(x) \sim \frac{1}{\nu} \frac{1}{(\text{Var}_\rho[\tilde{T}(x)])^2}. \quad (1.31)$$

Here  $\tau_{\text{Eig}}$  is the mean spacing in the homogeneous limit, whereas  $\tau_{\text{ref}}(x)$  acts as a local refractory period after strong localization.

**Time-production functional.** The relation between the auxiliary coordinate  $s$  (a reparametrization-invariant bookkeeping parameter; see Appendix B) and emergent time  $t$  is fixed by

$$\frac{dt}{ds} = \Gamma[\Psi], \quad t(s) = \int_0^s \Gamma[\Psi(s')] ds', \quad (1.32)$$

The lower integration bound is a convention fixing the origin of the auxiliary parameter  $s$ ; physical statements depend only on differences  $t(s_2) - t(s_1) = \int_{s_1}^{s_2} \Gamma[\Psi(s')] ds'$  and are invariant under  $s$ -translations.

The functional  $\Gamma[\Psi]$  is given by

$$\begin{aligned} \Gamma[\Psi] = \tau_{\text{Eig}} \bigg[ & \nu \int \frac{d^4x}{V_\xi} \text{Tr}[\tilde{T}^2(x) \rho \tilde{T}^2(x)] \\ & + \iint \frac{d^4x d^4y}{V_\xi^2} \gamma(x, y) C(\Psi; x, y) \text{Tr}[L_{xy}^\dagger L_{xy} \rho] \bigg]. \end{aligned} \quad (1.33)$$

The prefactor  $\tau_{\text{Eig}} = \nu^{-1}$  renders  $\Gamma$  dimensionless; by the law of large numbers,  $t(s) \approx \tau_{\text{Eig}} N_{\text{Eig}}(s)$ . For later use we isolate the bilocal term:

$$\Gamma_{\text{bi}}[\Psi(s)] = \iint \frac{d^4x d^4y}{V_\xi^2} \gamma(x, y) C(\Psi; x, y) \langle (\tilde{T}^2(x) - \tilde{T}^2(y))^2 \rangle_{\rho(s)}. \quad (1.34)$$

**Lemma 1.1** (Monotonicity of  $t$ ). *Let  $\rho(s) \geq 0$  with  $\text{Tr} \rho(s) = 1$ . Assume time-sector nondegeneracy (ND): either  $\exists x_0$  with  $\text{Tr}[\tilde{T}^2(x_0) \rho(s) \tilde{T}^2(x_0)] > 0$ , or  $\exists (x, y)$  with  $C[\rho(s); x, y] > 0$  and  $\text{Tr}[\rho(s) (\tilde{T}^2(x) - \tilde{T}^2(y))^2] > 0$ . With  $\nu > 0$ ,  $\gamma(x, y) \geq 0$ , and  $0 \leq C[\rho(s); x, y] \leq 1$ , one has  $\Gamma[\rho(s)] \geq 0$ , with  $\Gamma > 0$  under (ND); hence  $dt/ds = \Gamma[\rho(s)] \geq 0$  and  $t(s)$  is strictly increasing under (ND).*

**Lemma 1.2** (Eigtime activation on chronogenetic segments). *Under the standing integrability assumptions on  $\hat{T}(x)$  (finite fourth moments), assume nondegeneracy (ND) holds along a segment of the  $s$ -flow, so that  $\Gamma[\rho(s)] > 0$  on that segment (Lemma 1.1). Define the survival probability of having no eigtime event*



on a segment of length  $\delta > 0$  by

$$S_{s_0}(\delta) := \exp\left(-\int_{s_0}^{s_0+\delta} \frac{\Gamma[\rho(s')]}{\tau_{\text{Eig}}} ds'\right).$$

Then  $S_{s_0}(\delta) < 1$  on any chronogenetic segment (equivalently: eigentime events have non-zero intensity on every such segment). No segment is privileged: the statement is  $s$ -translation invariant.

**Lemma 1.3** (Contraction to equilibrium and  $\alpha = 2$ ). *Let  $\Phi_{s,\delta s}$  be the update–mix map over  $[s, s + \delta s]$ . In the Markov regime  $\mu = \tau_{\text{corr}}/\tau_{\text{Eig}} \ll 1$  there exist  $L \gg \xi$  and  $\varepsilon = \Theta(\mu) > 0$  such that*

$$\|\Phi_{s,\delta s}(\rho) - \Phi_{s,\delta s}(\sigma)\|_1 \leq (1 - \varepsilon) \|\rho - \sigma\|_1, \quad \forall \rho, \sigma.$$

**(Uniqueness and mixing)** *Since the state space is complete in trace norm, the strict contraction above implies (by the Banach fixed-point theorem) a unique fixed point  $\rho_\infty$  of  $\Phi_{s,\delta s}$ , and exponential convergence of iterates:*

$$\|\Phi_{s,\delta s}^n(\rho) - \rho_\infty\|_1 \leq (1 - \varepsilon)^n \|\rho - \rho_\infty\|_1, \quad \|\Phi_{s,\delta s}^n(\rho) - \Phi_{s,\delta s}^n(\sigma)\|_1 \leq (1 - \varepsilon)^n \|\rho - \sigma\|_1.$$

*This uniqueness/mixing mechanism is purely Markovian (Dobrushin-type) and does not rely on any KMS/detailed-balance/thermal structure. A Schauder–Tychonoff existence proof under constraints (without a contraction estimate) and an gives an independent (specialized) uniqueness route via primitivity and KMS spectral gap. With  $\varepsilon = \Theta(\mu)$  and the collective-stability law  $\tau_{\text{coh}}(N) \propto N^\alpha$ , the fixed-point balance  $\mu \sigma_\alpha = \mathcal{O}(1)$  uniquely selects  $\alpha = 2$ .*

*Sketch.* Finite interaction range  $\xi$  and the  $N^{-2}$  normalization imply bounded per-site influence  $\mathcal{O}(\mu)$  on blocks  $L \gg \xi$ . Standard Dobrushin-type arguments for finite-range Markov mixing then yield trace-norm contraction with  $\varepsilon = \Theta(\mu)$ , and the Banach fixed-point theorem gives uniqueness and exponential mixing. An independent uniqueness route via KMS/detailed-balance specialization exists in the near-horizon regime.  $\square$

**Self-consistent closure.** Lemma 1.1 ensures  $\Gamma[\rho] \geq 0$ , with  $\Gamma > 0$  under non-degeneracy (ND), so  $t(s)$  is non-decreasing in general and strictly increasing under (ND), defining an intrinsic arrow of time. No external clock is introduced at any stage: the statistics of eigentimes, constrained by WESH–Noether conservation and collective stability in the Markov window, fully determine the emergent

temporal order. Markovianity, Poissonian statistics, and the atemporal bootstrap architecture thus form a single logical package: the evolution depends only on the present state, carries no temporal memory, and is closed under self-consistency. The foundational circularity (WESH derived from Wheeler–DeWitt by constraints, and the WESH dynamics contracting back to the frozen Wheeler–DeWitt constraint in the  $G \rightarrow 0$  limit) reflects the self-consistency of the construction in a pre-geometric theory: time, geometry, and dynamics co-emerge from the same WESH completion of the timeless constraint.

**Remark 1.3** (Pre-geometric proximity and emergent causal structure). *The kernel  $\gamma(x, y)$  is defined via the pre-geometric proximity  $N_\xi$ , with no metric input. The WESH dynamics contracts to a unique fixed point  $\rho^*$  (Lemma 1.3). At  $\rho^*$ , the emergent metric  $g_{\mu\nu}^{(T)}$  is well-defined and Lorentzian on scales  $L \gg \xi$  (via gradient alignment at the fixed point). The rate envelope  $K_\xi$  decays exponentially beyond range  $\xi$ , so the dissipative kernel couples only pairs within one correlation length of each other. It follows that  $N_\xi$  and the causal domain of  $g_{\mu\nu}$  agree on all dynamically relevant pairs, up to corrections  $\mathcal{O}(\xi/L)$  controlled by the Markov parameter  $\mu = \tau_{\text{corr}}/\tau_{\text{Eig}} \ll 1$ . That a causal order determines the conformal geometry (Hawking, King & McCarthy, 1976; Malament, 1977) confirms that the pre-geometric proximity  $N_\xi$  is strictly weaker than a metric assumption. Metric causality is a consequence of the fixed point, not an input to its construction.*

## 1.8 Foundational consistency and parameter fixing

A necessary check on any dissipative extension of Wheeler–DeWitt is the behavior of the theory in limiting regimes. Here we verify the decoupling of gravitational effects ( $G \rightarrow 0$ ) and the consistency of fundamental scales; the thermodynamic scaling at horizons lies beyond the present scope.

**Finite memory (Born–Markov).**

$$\int_0^\infty dt \|\langle \mathcal{C}(t) \mathcal{C}(0) \rangle\| < \infty, \quad \tau_{\text{corr}} = \xi/c \ll \tau_{\text{Eig}} \equiv 1/\nu. \quad (1.35)$$

*Note.* We use  $\mathcal{C}(t)$  for the memory kernel to avoid clashes with the entanglement gate  $C(\Psi; x, y)$ .

**Compatibility with  $N^2$  stability (ratio scaling).**

$$\frac{\tau_{\text{corr}}}{\tau_{\text{Eig}}^{(\text{eff})}} \sim N^{-2} \quad (\text{bilocal } \gamma \propto N^{-2}, \tau_{\text{Eig}}^{(\text{eff})} \propto N^2). \quad (1.36)$$

**Fundamental scales.** The fundamental scales of the theory are anchored to Planck units:

$$\xi = \frac{\hbar}{m_T c} \simeq L_P, \quad \gamma_0 \simeq t_P^{-1}. \quad (1.37)$$

*Terminology.*  $m_T$  fixes the range  $\xi$  of the kernel used in  $\gamma(x, y)$ ; it is not an on-shell degree of freedom.

**Proposition 1.1** (Decoupling in the  $G \rightarrow 0$  limit). *Let  $\xi \simeq L_P \propto \sqrt{G}$  and  $\gamma_0 \simeq t_P^{-1} \propto G^{-1/2}$ , and write*

$$\gamma(x, y) = \frac{\gamma_0}{N^2} K_\xi(x - y) \mathbb{K}_{N_\xi}(x, y), \quad K_\xi \geq 0, \quad \|K_\xi\|_{L^1(\mathbb{R}^{1,3})} \sim V_\xi = \mathcal{O}(\xi^4).$$

*Define the shorthand*

$$\int \gamma \equiv \int d^4 y \gamma(x, y).$$

*Then*

$$\nu \propto \gamma_0 V_\xi \rightarrow 0, \quad \int \gamma = \frac{\gamma_0}{N^2} \|K_\xi\|_{L^1} \sim \frac{\gamma_0}{N^2} V_\xi \rightarrow 0,$$

*so for any bounded operator  $A$ ,*

$$\left\| \frac{d}{ds} \text{Tr}(A\rho) \right\|_{\text{diss}} \leq C_A (\nu + \int \gamma) \rightarrow 0.$$

*Proof sketch.*  $\|K_\xi\|_{L^1} \sim c \xi^4$  yields  $\nu, \int \gamma = \mathcal{O}(\gamma_0 \xi^4) \rightarrow 0$ ; standard Lindblad bounds give  $\|\mathcal{D}\| \leq \text{const} \cdot (\nu + \int \gamma) \rightarrow 0$ , hence strong convergence to the unitary group.

**Constraint continuity (The Frozen Limit).** Since the fundamental scales satisfy  $\xi \sim L_P \propto \sqrt{G}$  and  $\gamma_0 \sim t_P^{-1}$ , taking the formal limit  $G \rightarrow 0$  implies  $\nu \rightarrow 0$  and  $\int \gamma \rightarrow 0$ . In this limit, the dynamics undergoes a structural contraction:

- (i) **Vanishing dissipation.** The non-unitary terms in the master equation scale to zero, leaving a unitary flow generated solely by  $\hat{H}_{\text{eff}}$ .
- (ii) **Halting of temporal flow.** Crucially, the eigentime event rate  $\langle dN_{\text{Eig}}/ds \rangle$  (Eq. 1.28) vanishes as  $\nu \rightarrow 0$  and  $\int \gamma \rightarrow 0$  (Prop. 1.1): no eigentimes occur, and the arrow of time ceases to emerge.

**Physical interpretation.** The limit  $G \rightarrow 0$  enables us to recognize WESH and Wheeler–DeWitt as two regimes of the same fundamental constraint. When gravitational coupling is active, the constraint unfolds into a dissipative, dynamical structure that generates temporal flow; when gravity is turned off, both the production of time and the emergence of spacetime geometry halt simultaneously, and the theory continuously contracts back to the frozen, timeless sector  $\hat{H}_{\text{tot}}|\Psi\rangle = 0$ .

Physical time emerges as the output of a single dissipative resolution of the Wheeler–DeWitt constraint, through the eigentime bootstrap ( $dt/ds = \Gamma > 0$ ). The same pre-geometric algebraic constraints—CPT invariance, WESH–Noether charge conservation, and collective stability—also determine the infrared attractor of the WESH flow, from which spacetime geometry crystallizes through gradient alignment. Neither time nor geometry is logically prior; both derive from the same dissipative structure without postulating a background arena.

**Fixed-regime summary.** Combining the  $N^{-2}$  kernel scaling with the Markov condition  $\tau_{\text{corr}} \ll \tau_{\text{Eig}}$  yields a self-consistent fixed point:

$$\boxed{(\tau_{\text{coh}}(N) \propto N^2) \ \& \ (\mu := \frac{\tau_{\text{corr}}}{\tau_{\text{Eig}}} \ll 1) \iff \alpha = 2} \quad (1.38)$$

The bootstrap mechanism is a self-consistency requirement that constrains the dynamics of the theory, analogous to closure conditions successfully employed in conformal field theory to fix operator dimensions (Rattazzi et al., 2008). This is not circular; it constitutes a dynamical equilibrium condition: the parameters that govern dissipation ( $\gamma_0, \xi$ ) and the resulting statistical properties of eigentime events ( $\nu, \tau_{\text{coh}}$ ) must mutually adjust to maintain Markovian memory within a collective  $N^2$  stability window.

**Remark 1.4.** *The scaling statement in Eq. (1.38) is encoded by the bilocal part of the WESH generator (1.15) with finite-range kernel (1.20): the  $N^{-2}$  prefactor provides collective protection (per-site rate  $\Gamma_i \sim \gamma_0/N$ ), while the finite correlation length  $\xi$  fixes a state-independent correlation time  $\tau_{\text{corr}} \sim \xi/c$ . Together these imply the Markov window  $\mu := \tau_{\text{corr}}/\tau_{\text{Eig}} \ll 1$  in the large- $N$  regime (see the heuristics in Appendix C).*

Table 2 collects the fundamental parameters and scales of the construction.

**Table 2: Summary of fundamental parameters and scales in the QFTT–WESH formalism.** All quantities use natural units with  $\hbar = c = 1$  where applicable.

| Symbol               | Meaning                    | Dimensions     | Scale / Definition                | Eq.            |
|----------------------|----------------------------|----------------|-----------------------------------|----------------|
| $\xi$                | Corr. length               | [length]       | Planck ( $\simeq L_P$ )           | (1.37), (1.20) |
| $\gamma_0$           | WESH rate scale            | [time] $^{-1}$ | $\simeq t_P^{-1} \times \Theta_0$ | (1.37), (A.6)  |
| $N_{\text{Eig}}$     | eigentime count            | —              | via Eq. (1.28)                    | (1.28), (1.32) |
| $\tau_{\text{Eig}}$  | Mean eigentime spacing     | [time]         | $1/\nu$                           | (1.31)         |
| $\nu$                | Local eigentime rate       | [time] $^{-1}$ | matched (no explicit $N$ )        | (1.28), (A.5)  |
| $\tau_{\text{corr}}$ | Correlation time           | [time]         | $\xi/c$                           | (1.35)         |
| $\tau_{\text{coh}}$  | Coherence time             | [time]         | $\propto N^2$                     | (1.38)         |
| $\alpha$             | Stability exponent         | —              | 2                                 | (1.38)         |
| $\Gamma[\Psi]$       | Time-production functional | —              | $\geq 0$                          | (1.32), (1.33) |
| $C(\Psi; x, y)$      | Entanglement corr.         | —              | $[0, 1)$                          | (1.21)         |
| $L_{xy}$             | Bilocal jump operator      | —              | $\tilde{T}^2(x) - \tilde{T}^2(y)$ | (1.9), (1.13)  |
| $m_T$                | Kernel mass scale          | [mass]         | $\propto 1/\xi$                   | (1.37)         |

*Units Note.* With  $\tilde{T} = \hat{T}/\tau_s$  and normalized measures  $\int \frac{d^4x}{V_\xi}$  and  $\iint \frac{d^4x d^4y}{V_\xi^2}$ , each additive term in the generator carries units of inverse time. With  $\|K_\xi\|_{L^1(\mathbb{R}^{1,3})} \sim V_\xi = \mathcal{O}(\xi^4)$ , the integrated bilocal weight scales as

$$\int \gamma = \frac{\gamma_0}{N^2} \|K_\xi\|_{L^1} \sim \frac{\gamma_0}{N^2} V_\xi.$$

Using the normalized measures ensures consistent dimensions in Eq. (1.15). The time-production functional  $\Gamma[\Psi]$  is rendered dimensionless by the prefactor  $\tau_{\text{Eig}} = 1/\nu$  in Eq. (1.33), ensuring that the map  $dt/ds = \Gamma[\Psi]$  is dimensionally consistent.

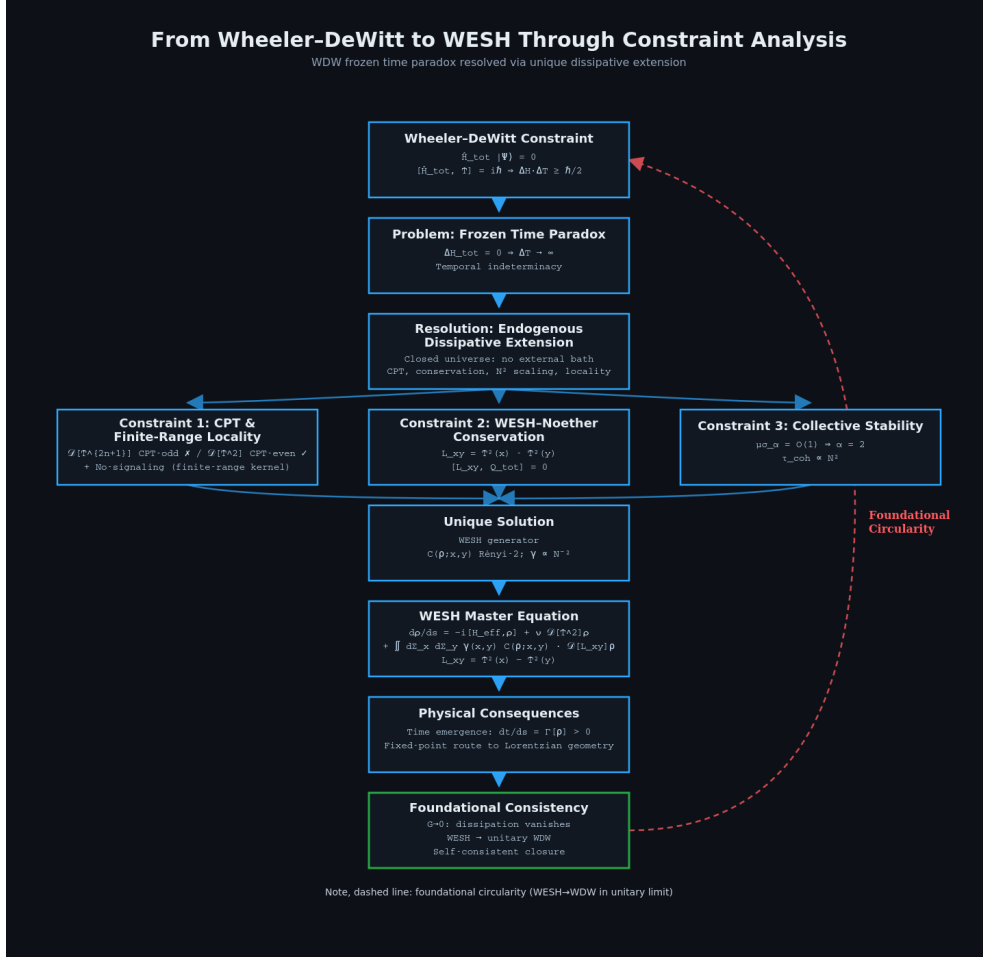


Figure 1.1: **WESH as the dissipative extension of Wheeler-DeWitt.** The frozen time paradox ( $\Delta H_{\text{tot}} = 0 \Rightarrow \Delta T \rightarrow \infty$ ) requires a GKSL dissipative structure. Three constraints (CPT symmetry with finite-range locality, WESH-Noether charge conservation, and collective  $N^2$  stability) determine the generator: a local channel  $\mathcal{D}[\hat{T}^2(x)]$  plus a bilocal channel  $\mathcal{D}[L_{xy}]$  weighted by Rényi-2 correlators. The dashed arrow indicates foundational closure: in the limit  $G \rightarrow 0$ , dissipation vanishes and WESH reduces to unitary Wheeler-DeWitt.

## 2 Key Theoretical Predictions

*Overview.* Building on the WESH–Noether structure developed in Sec. 1, we now isolate two quantitative signatures of the framework: (i) collective stability, expressed by the  $N$ –scaling of coherence, which acts as a structural discriminant of the bilocal WESH channel; and (ii) a  $\cos^2 \theta$  angular modulation in parity decay, which provides a diagnostic test of state/geometry factorization and finite-range connectivity. Both follow from the WESH master equation and its finite-range kernel, and they provide quantitative targets for the simulations and hardware tests of Sec. 3.

### 2.1 Collective stability and $N$ -scaling

The fundamental signature of WESH dynamics is the suppression of pairwise dissipative couplings as the system size grows. Unlike standard decoherence, where independent local errors accumulate linearly, the WESH interaction strength scales inversely with the square of the number of constituents:

$$\gamma_{ij}(N) = \frac{\gamma_0}{N^2}, \quad i \neq j. \quad (2.1)$$

The  $N^{-2}$  scaling emerges as a non-arbitrary requirement: maintaining the total collapse power finite in the thermodynamic limit ( $\sum \gamma_{ij} \sim \mathcal{O}(\gamma_0)$ ), ensuring that the effective rate per subsystem decreases as  $\Gamma_i \sim 1/N$ .

Consequently, we predict a **collective enhancement of coherence time** that defies standard local noise models:

$$\tau_{\text{coh}}(N) \propto N^2. \quad (2.2)$$

This quadratic protection is valid in the Markovian regime where the correlation time  $\tau_{\text{corr}}$  is small compared to the eigentime scale  $\tau_{\text{Eig}}$  (see Appendix C for the heuristic derivation).

### 2.2 Angular dependence: projection $\rightarrow$ geometry $\rightarrow$ state

A second, diagnostic signature connects the detector geometry to the prepared quantum state. The decay rate of parity oscillations is predicted to follow a robust

angular modulation:

$$\Gamma_{\text{dec}}(\theta) = \bar{\Gamma}(1 + \varepsilon \cos^2 \theta), \quad \varepsilon = \frac{a_2}{a_0} \frac{G_2}{G_0}. \quad (2.3)$$

Here  $\varepsilon$  factorizes into an intrinsic state anisotropy ( $a_2/a_0$ ) and a device-specific geometric factor ( $G_2/G_0$ ). The  $\cos^2 \theta$  form follows from the WESH master equation through a chain of four steps, which we summarize here (the full derivation with all intermediate expressions is available in the supplementary material).

**Step 1: Dynamical origin.** In the Heisenberg picture, the WESH bilocal channel governs the parity evolution through  $\frac{d}{ds} \langle \hat{\Pi}(\theta) \rangle_{\text{WESH}} = \iint \gamma(x, y) C(\Psi; x, y) \langle L_{xy}^\dagger \hat{\Pi} L_{xy} - \frac{1}{2} \{L_{xy}^\dagger L_{xy}, \hat{\Pi}\} \rangle$ . In the echo-narrowband regime at fixed  $(T, n)$ , this reduces to a state-dependent scalar rescaling:  $\frac{d}{ds} \langle \hat{\Pi} \rangle = -R_\Pi M(\Psi) \langle \hat{\Pi} \rangle$ , where  $R_\Pi$  is angle-independent and the modulator  $M(\Psi) \propto \bar{C}(\Psi) \cos^2 \vartheta$  encodes the entanglement correlator and the structural overlap of the prepared state. This factorization, not the geometry, fixes the existence of the  $\cos^2$  harmonic.

**Step 2: Gradient expansion.** Within the Markov window, the bilocal integral reduces to a 3D spatial average at fixed  $s$ . A second-order Taylor expansion of  $\hat{T}^2(y)$  around  $x$  yields  $(\hat{T}^2(x) - \hat{T}^2(y))^2 \approx r^2(\hat{\mathbf{r}} \cdot \nabla \hat{T}^2)^2$ , which links the scalar difference to a direction-selective form.

**Step 3: Measurement projection.** The rotated-parity observable  $\hat{\Pi}(\theta) = (\cos \theta \sigma_z + \sin \theta \sigma_x)^{\otimes N}$  acts as a geometric projector onto pair orientations:  $(\hat{\mathbf{m}}(\theta) \cdot \hat{\mathbf{r}}_{ij})^2 = \cos^2(\theta - \alpha_{ij})$ . The  $\cos^2 \theta$  harmonic emerges from the interplay of this projector with the pair-orientation statistics of the device layout, encoded in the lattice sums  $G_0 = \sum_k w_k$  and  $G_2 = \sum_k w_k \cos 2\alpha_k$ .

**Step 4: Gate anisotropy.** For GHZ states prepared with directional bias  $\hat{\mathbf{n}}$ , the entanglement gate admits an even expansion  $C(\Psi; \hat{\mathbf{r}}) = a_0 + a_2(\hat{\mathbf{r}} \cdot \hat{\mathbf{n}})^2 + \mathcal{O}(a_4)$ , with  $a_0 > 0$  and  $|a_2| \ll a_0$ . Inserting this into the lattice-weighted sum yields  $\varepsilon = (a_2/a_0)(G_2/G_0)$ . For W-type states,  $a_2^{(W)} < 0$ , predicting anti-modulation ( $\varepsilon_W < 0$ ).

The key point is that each step in the derivation is anchored to the WESH dissipator structure: the  $\cos^2 \theta$  form is a consequence of the quadratic jump operators  $L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)$ , the finite-range kernel, and the parity readout, not of any ad hoc geometric assumption. The device factor  $G_2/G_0$  can be computed independently from the chip layout, allowing an inversion  $a_2/a_0 = \varepsilon/(G_2/G_0)$



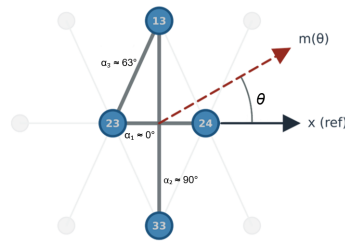
that extracts the intrinsic state anisotropy from the measured modulation amplitude. GHZ states exhibit positive modulation ( $\varepsilon > 0$ ), while W-states show anti-modulation ( $\varepsilon < 0$ ), providing a state-class control that isotropic noise models cannot reproduce.

#### WESH Angular Law — Geometric Projection

Pair projection:  $\cos^2(\theta - \alpha_i)$

$\varepsilon = (a_2/a_0) \cdot (G_2/G_0)$

Appendix F — Eq. (F.11), (F.12)



Minimal geometry: three representative pair orientations  $\alpha_i$ . Measurement axis  $m(\theta)$  filters pairs via  $\cos^2(\theta - \alpha_i)$ .

For symmetric heavy-hex subsets, the  $\sin 2\theta$  component cancels by reflection symmetry, leaving the  $\cos^2\theta$  harmonic.

Figure 2.1

## 2.3 Discrimination from conventional decoherence models

The predictions of Sec. 2.1–2.2 are informative only insofar as they can be *discriminated* from conventional (non-WESH) decoherence mechanisms. Here we (i) state an explicit null class of models, (ii) isolate the two WESH-specific discriminants already derived in the framework (collective  $N$ -scaling and state–geometry factorization of the angular law), and (iii) summarize falsification criteria and the minimal control program needed to exclude device-driven explanations.

**Status of the collision model.** The collision model used in Sec. 3 is a deliberately reduced *surrogate*: it does not attempt to simulate the full WESH master equation (e.g. the full finite-range-kernel structure or the pre-geometric WESH–Noether constraint), but isolates the specific ingredient tested there, namely the inverse scaling of the effective couplings implied by Eq. (2.1), in a tractable Markovian dynamics with an explicit local benchmark. In this controlled setting, extracting effective decay rates yields a clear pre-asymptotic power law  $\gamma(N) \propto N^{\alpha_\gamma}$

with  $\alpha_\gamma \simeq -1.80$  over  $N = 2\text{--}16$  (while the matched local benchmark returns  $\alpha_\gamma = -1$ ), trending toward the WESH asymptote  $\alpha_\gamma = -2$  and already cleanly separated from local scaling. The collision model therefore serves as a *quantitative consistency check* (numerical proof-of-concept) for the collective-stability mechanism and for the scaling-inference pipeline; it complements, rather than replaces, the hardware evidence. Empirical support is instead sought in the hardware signatures of Sec. 3, where the angular modulation and the entanglement-linked protection are observed under gate-matched controls on real processors.

**Null class of models (non-WESH).** We take as baseline any model in which decoherence is fully accounted for by *state-independent hardware noise* (possibly including spatial correlations) plus control imperfections. Concretely, this includes: (i) independent local Markovian channels (dephasing, depolarizing, amplitude damping) with per-qubit rates not decreasing with system size, and (ii) device-level correlated noise that may be anisotropic in space but remains *independent of the prepared entanglement class* (GHZ vs W vs product), once gate depth and readout are matched. Under this null class, any observed advantage of an entangled preparation must be attributable either to circuit-level differences or to geometry-only correlations, not to a bilocal, state-gated channel.

**Structural discriminant: inverse pair scaling and collective stability.** Within the WESH master-equation structure used in Sec. 2.1, the suppression of pairwise dissipative couplings,

$$\gamma_{ij}(N) = \gamma_0/N^2, \quad i \neq j$$

(Eq. (2.1)) is the condition that keeps the total bilocal collapse power finite when the number of pairs grows with  $N$ . In the regime in which the reduced description applies (Markov window as specified in the framework), this yields the collective stability enhancement

$$\tau_{\text{coh}}(N) \propto N^2,$$

(Eq. (2.2)). By contrast, in the null class above, increasing  $N$  does not generically increase coherence time for genuinely entangled states beyond trivial control effects: independent local error accumulation is constant at best and typically degrades with  $N$ , and correlated noise can at most change the *rate of degradation* unless one engineers a symmetry-protected subspace (addressed below). Therefore, observing sustained superlinear enhancement in entangled coherence with growing  $N$  is a *structural* rejection of the baseline class, while the absence of any superlinear trend across accessible  $N$  places a quantitative upper bound on the WESH coupling and, if persistent, would falsify the collective-stability prediction as stated.

**Diagnostic discriminant: factorized, state-dependent angular modulation.**

The angular law

$$\Gamma_{\text{dec}}(\theta) = \bar{\Gamma}(1 + \varepsilon \cos^2 \theta), \quad \varepsilon = \frac{a_2}{a_0} \frac{G_2}{G_0},$$

(Eq. (2.3)) is derived in Sec. 2.2 from (i) rotated-parity projection, (ii) lattice-weighted geometry, and (iii) weak even gate anisotropy (full intermediate expressions in the supplementary material). Importantly, the framework predicts not merely a  $\cos^2 \theta$  harmonic, but a *factorization*:  $G_2/G_0$  is device/ layout-specific and state-independent, while  $a_2/a_0$  is an intrinsic state-class parameter controlling the *sign* and magnitude of  $\varepsilon$ . A purely device-driven anisotropy can indeed generate a  $\cos^2 \theta$  modulation in parity decay, so the harmonic *alone* is not sufficient for discrimination. The diagnostic power arises from the *state-coupled* structure: when geometry and readout are held fixed, the framework predicts a sign reversal of the modulation between GHZ-type preparations ( $\varepsilon > 0$ ) and W-type controls ( $\varepsilon < 0$ ), as already used as a control in Sec. 3.

**Minimal control set implied by the framework.** Discrimination requires treating the WESH signature as a *joint constraint* rather than a single effect:

1. **Gate-matched entanglement control (Sec. 3).** The fake-GHZ procedure isolates entanglement from gate depth: if the observed protection were a circuit artifact, fake-GHZ would reproduce it; instead, the separation between GHZ and product-like controls is used as the operational discriminator.
2. **State swap at fixed geometry (Sec. 3).** Replacing GHZ by a W-state while keeping the same device subset and readout targets the state-coupled nature of  $\varepsilon$ .
3. **Cross-platform reproducibility (Sec. 3).** Reproducing the angular signature across distinct hardware stacks (IBM Eagle vs Rigetti Ankaa-3) reduces the plausibility of vendor-specific correlated noise as a complete explanation.
4. **Sequence/band robustness (Sec. 3).** Protection that persists across the spectroscopy band and across CPMG depths constrains interpretations in which the effect is primarily a dynamical-decoupling filter artifact.

The angular derivation (Sec. 2.2) further clarifies that an angle-independent baseline (hardware noise) can coexist with the modulation without generating it; hence, controls must target the *state dependence* of the modulation amplitude rather than the existence of a baseline rate.

| Signature                    | WESH (this work)                                                       | State-independent hardware noise (null class)                 | Standard alternatives                    |
|------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------|
| Angular modulation           | $\cos^2 \theta$ with factorization and GHZ/W sign flip                 | possible device-tied anisotropy, no state-class sign reversal | fixed pattern from bath / layout         |
| Scaling with $N$             | $\tau_{\text{coh}} \propto N^2$ (collective stability)                 | $\tau_{\text{coh}} = \mathcal{O}(1)$ or degrading with $N$    | model-dependent, typically sub-quadratic |
| Gate-matched GHZ vs fake-GHZ | protection persists under gate-matched control                         | differences attributable to overhead only                     | depends on encoding/implementation       |
| GHZ vs W modulation sign     | $\varepsilon_{\text{GHZ}} > 0$ , $\varepsilon_W < 0$ at fixed geometry | no systematic dependence on entanglement class                | no generic sign flip at fixed layout     |

**Table 3: Schematic discrimination logic.** The WESH framework is constrained by a *joint* signature set: collective  $N$ -scaling, factorized angular modulation, and state-dependent sign reversal at fixed geometry. Conventional noise mechanisms may reproduce individual rows but must account for the full pattern simultaneously.

**Alternative explanations and what they must reproduce.** The following conventional mechanisms can mimic *parts* of the observed phenomenology, but are constrained by the full signature set above:

1. **Spatially correlated Markovian noise (device anisotropy).** This can generate a  $\cos^2 \theta$  harmonic if correlations define a preferred direction. To explain the WESH pattern it must additionally reproduce (at fixed geometry) the GHZ/W sign reversal and the gate-matched GHZ vs fake-GHZ separation, rather than producing a modulation that tracks only device orientation.
2. **Non-Markovian (colored) noise under dynamical decoupling.** Such noise can produce strong sequence-dependent changes in apparent decay rates. To mimic the WESH claims it must yield a state-class dependent modulation (including sign reversal) that remains consistent across CPMG depths and across the tested frequency band.
3. **Decoherence-free subspace / collective-noise protection.** DFS effects

require an approximate permutation-symmetric coupling structure not naturally realized on fixed-geometry superconducting chips with inhomogeneous couplings. Moreover, DFS protection is highly state-selective; to replace the WESH explanation it must account simultaneously for the entanglement-specific protection (gate-matched control) *and* the state-coupled angular signature derived in Sec. 2.2.

**Falsification criteria.** Within the present formulation, the framework is constrained or ruled out by any of the following:

1. **No collective enhancement:** entangled coherence does not show a superlinear improvement with  $N$  under gate-matched conditions as hardware capabilities extend to larger multipartite registers.
2. **No state-coupled angular signature:** the  $\cos^2 \theta$  modulation is absent on entangled states, or appears identically on product/fake-GHZ controls (indicating a device artifact rather than a state-coupled effect).
3. **No GHZ/W sign reversal:** the modulation amplitude does not change sign under a GHZ  $\leftrightarrow$  W state swap at fixed geometry and readout.
4. **Protection compatible with overhead only:** the observed protection ratio remains consistent with gate-depth/readout overhead alone in the gate-matched control protocol.

**Experimental intent and topology awareness.** The simulations and hardware experiments reported in Sec. 3 were designed with falsification intent. The collision model implements the  $1/N^2$  coupling exactly and extracts scaling exponents from the output; the resulting  $\alpha_\gamma \simeq -1.80$  is a measured outcome. Had the exponent returned  $\alpha_\gamma \simeq -1$  (local scaling), the collective-stability prediction would have been ruled out at the simulation level.

The hardware experiments were equally agnostic. The  $\cos^2 \theta$  modulation and the GHZ/product protection gap are extracted from raw data under gate-matched controls; neither the circuit design nor the readout protocol encodes WESH-specific structure. The observed signatures are consistent with the predictions, but we are aware that the heavy-hex topology of IBM Eagle and the square-octagon lattice of Rigetti Ankaa-3 introduce geometry-dependent correlated noise that could, in principle, contribute to angular modulation independently of fundamental physics. For this reason, the falsification criteria above include state-class controls (GHZ vs W sign reversal, fake-GHZ null test) that are insensitive to fixed device geometry: a purely layout-driven artifact would not reverse sign under a state swap at fixed hardware geometry.

A definitive test of the geometric origin of the angular signature requires repeating the protocol on architectures with qualitatively different connectivity, in particular trapped-ion linear chains or reconfigurable neutral-atom arrays, where the pair-orientation statistics differ fundamentally from the heavy-hex lattice. Current data (Sec. 3) are consistent with the angular and protection signatures for  $N = 3, 4$  and show degradation for  $N \geq 5$ , defining the present sensitivity frontier on NISQ processors. Decisive tests of the collective-stability discriminant require higher-fidelity multipartite entanglement at larger  $N$ .

### 3 Experimental Validation

We test these predictions through a dual approach combining controlled classical simulations and quantum hardware experiments. The collision model constitutes a direct test of the collective-stability sector of the WESH master equation: the  $1/N^2$  coupling is implemented exactly, and the resulting scaling is a prediction, not an input. The hardware experiments provide a complementary test in a physical quantum system where the noise channels are not designed to match WESH but are observed to produce consistent phenomenology.

First, we employ a pre-asymptotic collision model on CPUs to isolate the collective stability mechanism. This model deliberately implements inverse couplings in a tractable regime ( $N \leq 16$ ) to verify the scaling laws without the overhead of full quantum simulation.

Second, we execute production runs on superconducting quantum processors (IBM Eagle and Rigetti Ankaa-3). These experiments probe the angular dependence and protection gaps in a physical setting, leveraging large-scale statistics ( $\sim 3 \times 10^6$  shots) to extract signals from the NISQ noise floor.

All datasets and analysis scripts are available in the companion repository:

<https://github.com/Luca-Casagrande/QFTT-WESH-1>.

#### 3.1 Numerical Evidence: Collective Stability

The collision model simulations are consistent with the non-standard scaling predicted by Eq. (2.1). Figure 3.1 compares WESH, standard decoherence, and pure WDW baselines across system sizes  $N = 3-9$ .

WESH vs Standard Decoherence vs Pure WDW - Analisi Completa

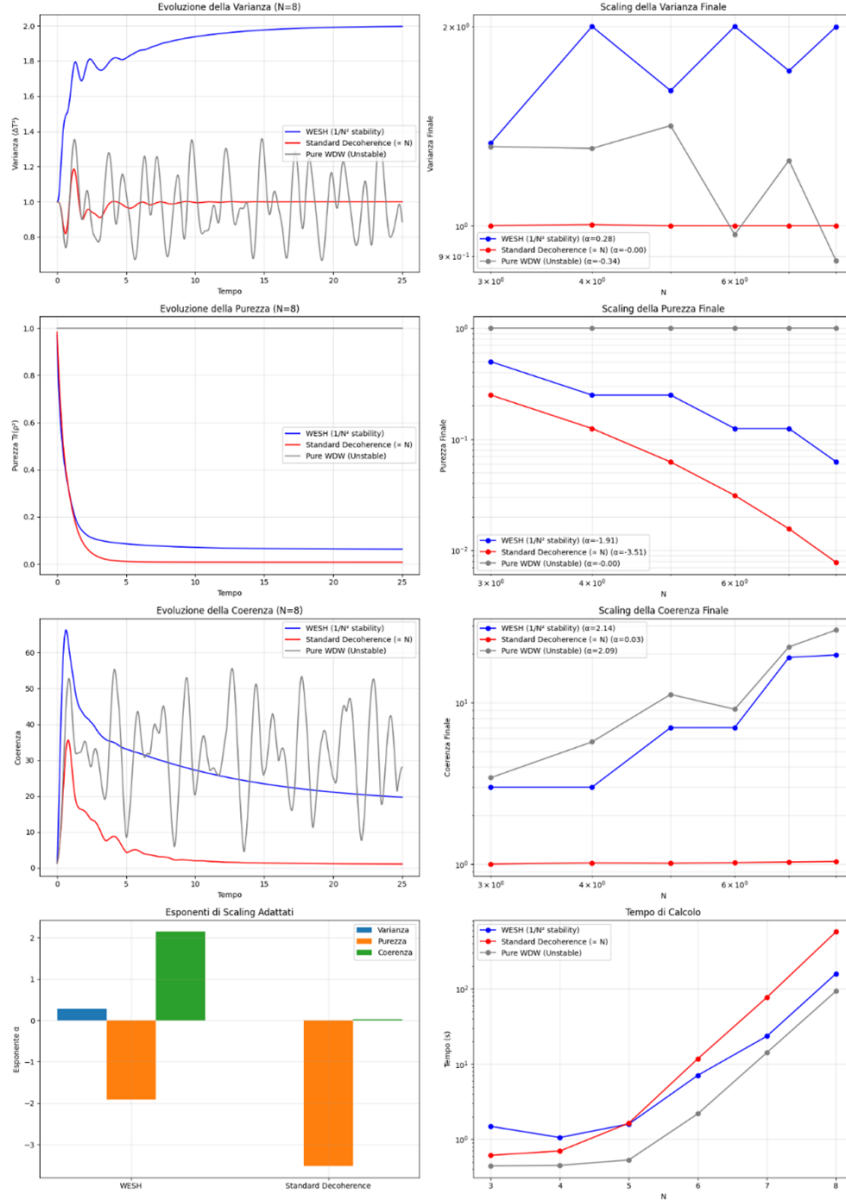


Figure 3.1: **Purity and coherence scaling** ( $N = 3-9$ ). Comparison of WESH (collective jump  $L \propto T/N$ ), Standard (local jumps  $\sigma_x$ ), and Pure WDW base-lines. **Left:** Time traces at  $N = 9$ . WESH maintains significantly higher purity and coherence. **Right:** Log-log scaling. WESH coherence grows super-linearly ( $\alpha_C \approx 2.5$ ), supporting the collective stability hypothesis. Standard decoherence shows no such enhancement.

The robustness of this scaling under different coupling profiles is shown in Figure 3.2.

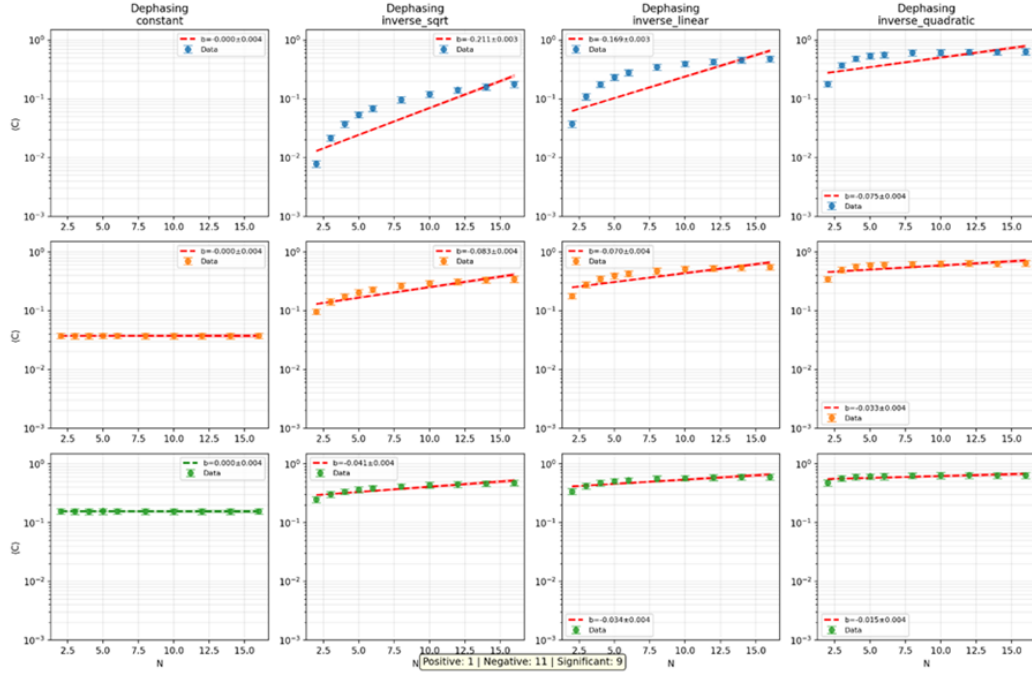


Figure 3.2: **Robustness of protection.** Final coherence  $C(N)$  for various noise channels and coupling profiles. Only inverse scaling ( $1/N^k$ ) enables coherence preservation (rising trends), whereas constant coupling leads to rapid decay (flat/dropping trends).

Extending the analysis to  $N = 16$  (Figures 3.3 and 3.4), the rate scaling exponent is extracted and compared against a local baseline.



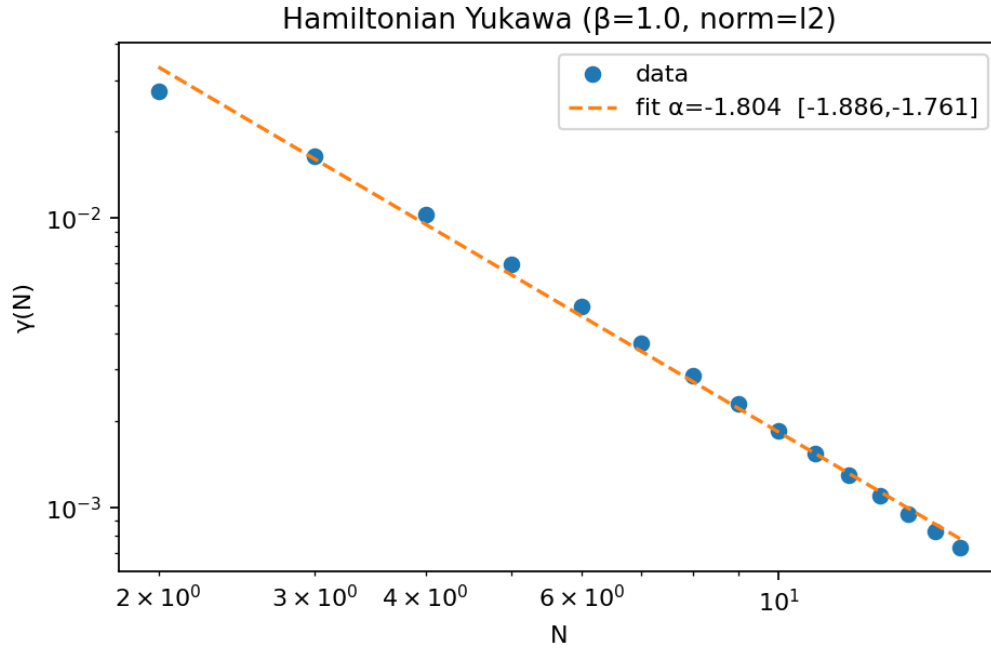


Figure 3.3: **Pre-asymptotic rate scaling.** Log-log regression yields  $\gamma(N) \propto N^{-1.80}$ , consistent with an approach to the asymptotic  $N^{-2}$  WESH prediction.

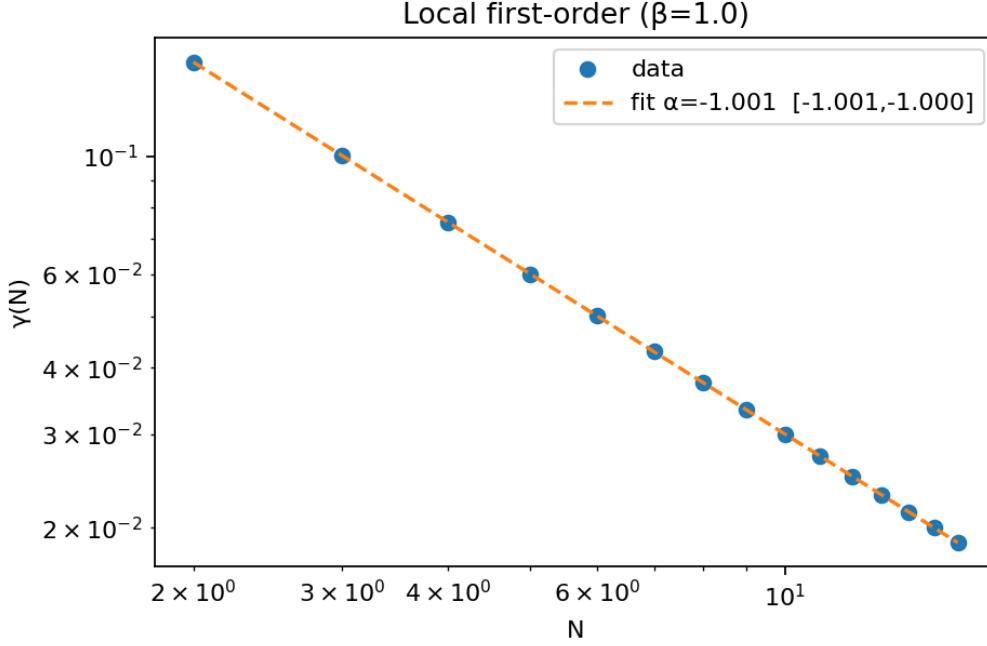


Figure 3.4: **Local baseline check.** The same analysis on a local first-order model yields  $\alpha = -1.00$ , validating the sensitivity of the metric.

### 3.2 Hardware Evidence: Angular Law and Protection

On quantum hardware, we focus on the angular signature (Eq. (2.3)). Experiments on IBM Eagle ( $N = 3, 4$ ) reveal a clear  $\cos^2 \theta$  modulation of the parity decay rate. Figure 3.5 shows a high-quality fit ( $R^2 = 0.947$ ) with modulation amplitude  $\varepsilon \approx 0.31$ . Crucially, preparing a W-state reverses the sign of the effect (anti-modulation), confirming the state-dependence predicted by the theory.

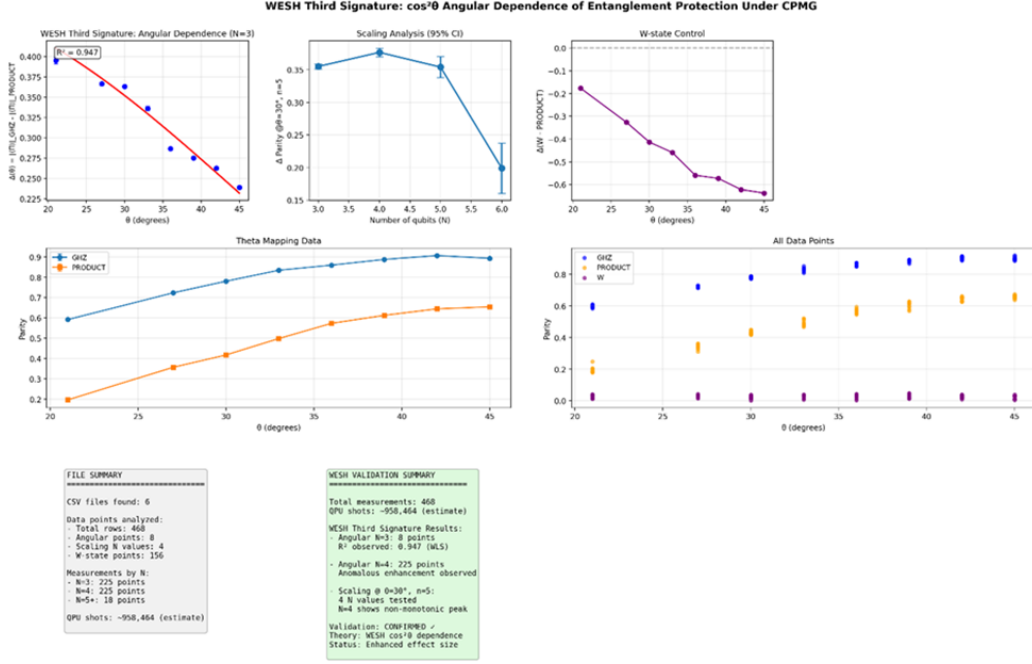


Figure 3.5: **Angular dependence on IBM Eagle. Top Left:** Decay rate modulation follows the predicted  $\cos^2 \theta$  law ( $R^2 = 0.947$ ). **Top Right:** W-state control shows the predicted sign reversal (anti-modulation).

To rule out gate-overhead artifacts, we performed a controlled comparison (Figure 3.6) between GHZ states and "fake-GHZ" states (entanglement created and immediately undone). The GHZ state retains significantly higher parity, with a decay rate ratio  $\Gamma_{\text{PROD}}/\Gamma_{\text{GHZ}} \approx 2.6$ , far exceeding the factor attributable to gate depth alone ( $\sim 1.2$ ).

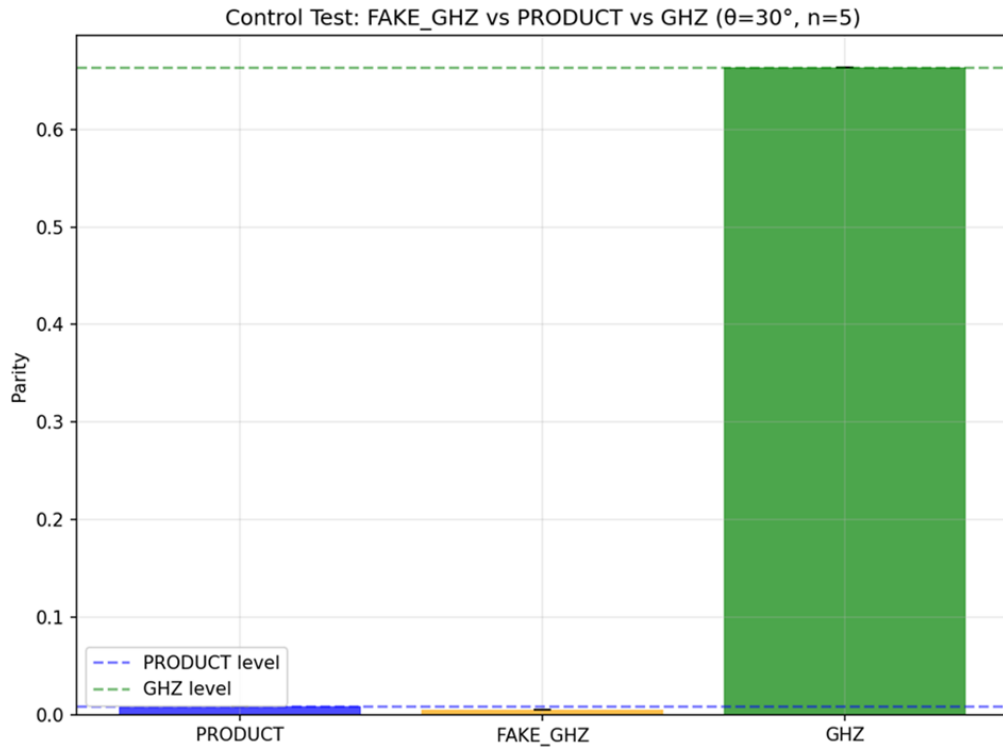


Figure 3.6: **Gate-matched control.** The "Fake-GHZ" (disentangled) behaves like the product state, while the true GHZ state maintains high parity. This isolates entanglement as the protective resource.

Comprehensive data from 1.9M shots (Figure 3.7) confirms this separation across the frequency spectrum. The distribution of parity outcomes for GHZ states is bi-modally separated from product states (Figure 3.8), providing model-independent evidence of entanglement-enhanced stability.

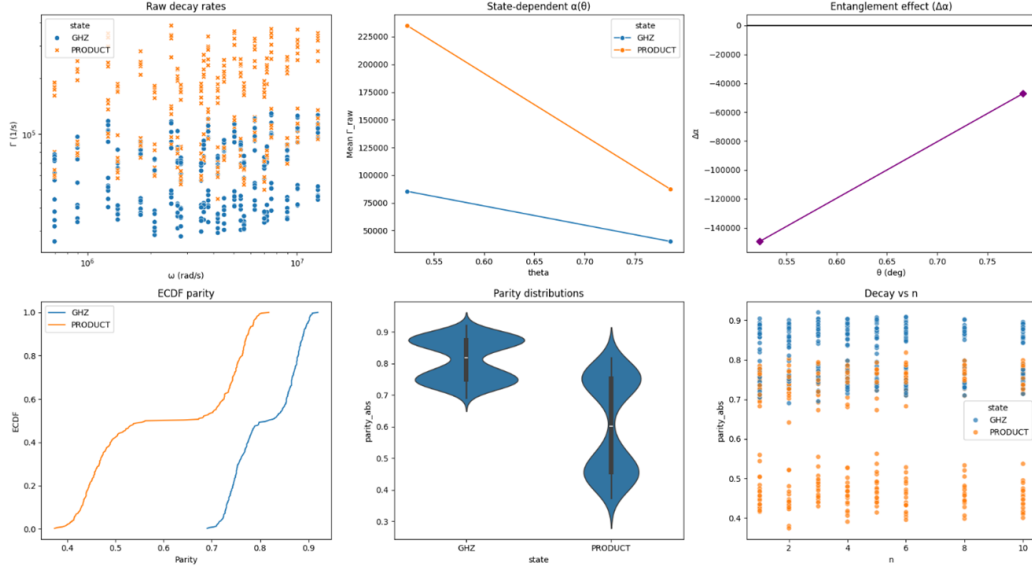


Figure 3.7: **Spectral consistency.** (a) Raw decay rates show GHZ (blue) consistently lower than product states (orange) across the 0.1–2.0 MHz band. (d) Protection persists across varying CPMG depths.

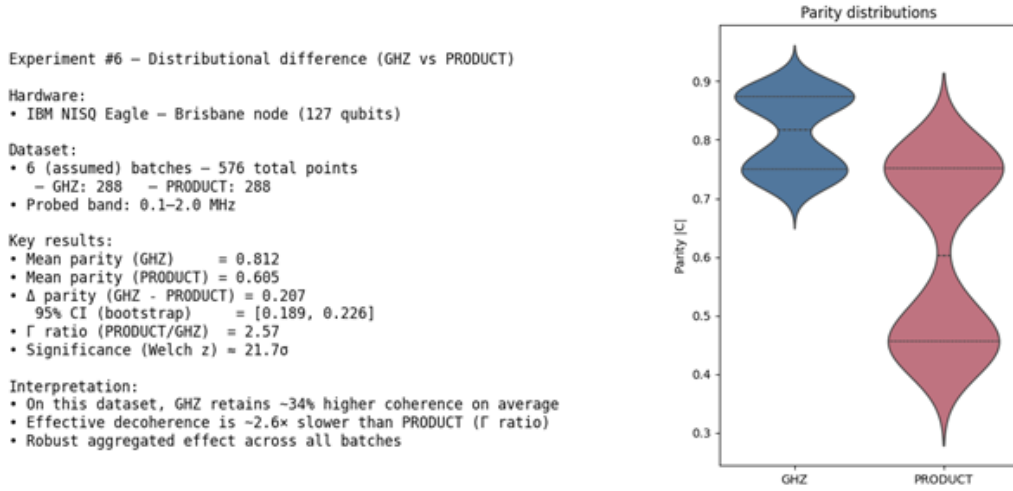


Figure 3.8: **Distributional separation.** Violin plots of parity outcomes ( $N = 576$ ) show a statistically significant gap ( $21\sigma$ ) between GHZ (blue, protected) and product states (red, decaying).

*Note.* In this dataset, multipartite entanglement is observed to suppress decoherence, as quantified by the  $\sim 2.6\times$  slower decay of the GHZ state. This protective effect, measured on a 127-qubit superconducting processor, highlights entan-

glement not only as a computational resource but also as a potential stabilizing mechanism for quantum information in near-term hardware, and merits further investigation independently of any specific theoretical framework.

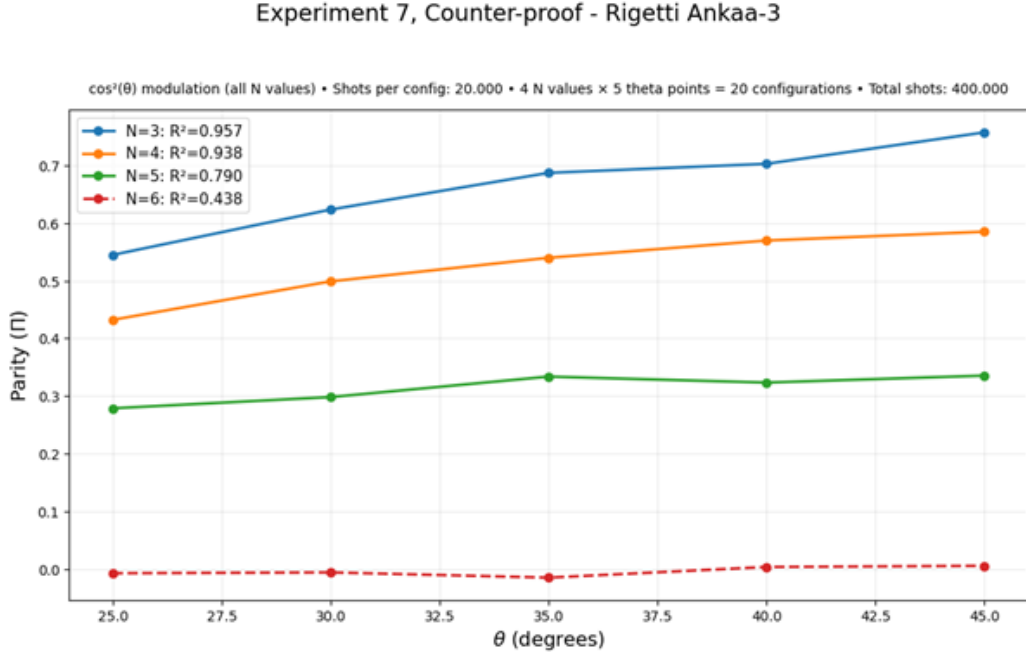


Figure 3.9: **Cross-platform check (Rigetti Ankaa-3).** The angular law is reproduced for  $N = 3, 4$ . Signal-to-noise ratio degrades for  $N \geq 5$ , defining the current hardware sensitivity limit.

| Metric                                   | WESH (Collision Model)           | Standard Noise | Pure WDW |
|------------------------------------------|----------------------------------|----------------|----------|
| Coherence Scaling ( $\alpha_C$ )         | +2.498 ( <b>protected</b> )      | +0.036 (flat)  | —        |
| Rate Scaling ( $\alpha_\gamma$ )         | −1.804 ( $N^{-2}$ <b>trend</b> ) | −1.001 (local) | —        |
| Purity Scaling ( $\alpha_{\text{pur}}$ ) | −1.909                           | −3.772         | $\sim 0$ |
| Thermodynamic Arrow                      | <b>Yes</b>                       | Yes            | No       |

**Table 4: Scaling metric comparison.** The WESH model uniquely combines a thermodynamic arrow of time with collective protection of coherence, distinct from both unitary evolution (Pure WDW) and standard local decoherence.

## Epilogue

The absence of time from the very equation that should describe its origin has long stood as one of the most compelling anomalies in fundamental physics. This work has attempted to resolve it from within: no external bath or clock, no background arena, only the internal resources of a closed quantum universe governed by the Wheeler–DeWitt constraint. With all the rigour at our disposal, we have sought to capture the beauty of a self-generating mechanism: one in which time is not assumed but produced by the very dynamics it enables.

The WESH master equation emerges as the natural dissipative completion of Wheeler–DeWitt: a minimal GKSL structure in which entanglement simultaneously carries statistical symmetry and drives the emergence of physical time through eigentime collapse events. No extra dimensions, supersymmetric partners, discrete spacetime atoms, or ad hoc chronon particles have been introduced; the construction relies entirely on the kinematics of quantum theory, the canonical constraints of General Relativity, and the structure of quantum entanglement. In particular, two structural keystones work in tandem: WESH–Noether conservation, enforced at the operator level before any background metric exists, and the bootstrap closure, through which the geometry necessary to define  $\hat{T}$  is generated by the field’s own dynamics. The various elements of the construction (CPT-even quadratic dissipators, the bilocal difference channel, the entanglement gate, the finite-range kernel, the  $N^{-2}$  collective normalization) emerge as the joint consequence of a small set of constraints applied to a closed pre-geometric universe; the parameters  $\nu$ ,  $k$ ,  $\alpha$  are structural constants locked by self-consistency.

Much remains to be done. The experimental programme is in its early stages: the collision-model simulations and the NISQ hardware campaigns reported here are consistent with the predicted  $N^2$  scaling and  $\cos^2 \theta$  angular law, but decisive tests require larger entangled registers and architectures with qualitatively different connectivity. We are conscious of these limitations and eager to extend the empirical contact.

Several theoretical directions are already under development and will appear in forthcoming work. The infrared fixed-point structure of the WESH dynamics provides a route to emergent Lorentzian geometry and the recovery of Einstein dynamics through gradient alignment and hidden-sector cancellation. The near-horizon regime yields Bekenstein–Hawking entropy from the same algebraic structure, without thermal ansatz. And the broader mathematical class to which WESH belongs (state-dependent quantum intensity processes with self-interacting Poisson structure, QSIPS) opens a line of investigation into the symmetry content of the bootstrap loop itself, with potential applications beyond quantum gravity.

Limitations remain. Near singularities, at trans-Planckian scales, or in regimes where the Markov condition  $\tau_{\text{corr}} \ll \tau_{\text{Eig}}$  fails, the assumptions underlying WESH cease to apply. What we can claim at this stage is a degree of internal coherence, between the algebraic constraints that select the generator, the dissipative dynamics that produce time, the experimental signatures that test the construction, and the fixed-point structure that connects to gravity, that invites further development.

## Acknowledgments

While the conceptual framework, methodology, and research direction remained with the author throughout, the work also relied on artificial intelligence tools in a cross-inferencing multi-AI workflow. The primary tools employed were those of OpenAI, Anthropic, and Google.

I dedicate this work with gratitude to my wife Margie, to the little Elvia, and to Kaïra.

## Appendix A: Flat-space realization of the kernel and the scales $\xi$ and $\gamma_0$

**Setup.** This appendix provides an explicit flat-space realization of the kernel defined abstractly in Sec. 1. The Minkowski background  $\eta_{\mu\nu}$  serves as an illustrative arena to fix dimensions and verify the scaling structure; the fundamental definition of  $\gamma(x, y)$  rests on the pre-geometric admissibility projector  $\mathbb{K}_{N_\xi}$  of Sec. 1 and does not require a background metric. Throughout this appendix we set  $\eta = \text{diag}(-, +, +, +)$  and  $\square = \eta^{\mu\nu} \partial_\mu \partial_\nu$ .

**A.1 Minimal mediator and retarded propagator.** We introduce a scalar mediator  $\chi(x)$  with Klein–Gordon dynamics on the pre-chronogenesis sector:

$$S_\chi = \int d^4x \left[ \frac{1}{2} (\partial_\mu \chi) (\partial^\mu \chi) - \frac{1}{2} m_T^2 \chi^2 \right]. \quad (\text{A.1})$$

The retarded Green function  $\Delta_R$  solves

$$(\square + m_T^2) \Delta_R(z) = \delta^{(4)}(z), \quad \text{supp } \Delta_R \subseteq \{z^0 \geq 0, z^2 \leq 0\}. \quad (\text{A.2})$$



The correlation length associated with the mediator mass is the Compton scale

$$\xi = \frac{\hbar}{m_T c}. \quad (\text{A.3})$$

**A.2 Positive finite-range envelope and correlation four-volume.** The retarded propagator  $\Delta_R$  fixes the *support* (future light cone) and the range  $\xi = \hbar/(m_T c)$ , but it is not used directly as a nonnegative  $L^1$  weight. In this flat-space realization, we adopt a positive  $L^1$  envelope  $K_\xi$  of range  $\xi$ , e.g.

$$K_\xi(z) = \exp\left(-\frac{z^0}{\xi}\right) \Theta(z^0) \Theta(-z^2), \quad (\eta = \text{diag}(-, +, +, +)), \quad (\text{A.4})$$

The dependence on  $z^0$  and  $z^2$  is specific to this Minkowski illustration; the pre-geometric definition (Sec. 1) requires only that  $K_\xi$  be positive,  $L^1$ -integrable, and supported on the finite-range neighbourhood  $N_\xi$ . Note that a positive  $L^1$ -integrable envelope with causal support necessarily breaks boost invariance (the future light cone has infinite Lorentz-invariant volume); the  $z^0$ -dependence above reflects this structural requirement, not a frame choice.

Its  $L^1$  norm defines the correlation four-volume

$$V_\xi := \int d^4 z K_\xi(z) = \mathcal{O}(\xi^4). \quad (\text{A.4}')$$

With the normalized measures of Eq. (1.14), this  $V_\xi$  is absorbed at the level of the generator. (If desired one may also define the normalized kernel  $\bar{K}_\xi := K_\xi/V_\xi$  so that  $\int d^4 z \bar{K}_\xi = 1$ .)

**A.3 Collapse kernel and dimensions.** The collapse kernel of the main text reads  $\gamma(x, y) = \frac{\gamma_0}{N^2} K_\xi(x - y) \mathbb{K}_{N_\xi}(x, y)$  (Eq. (1.20)). With  $K_\xi$  chosen positive and  $L^1$ -integrable,  $\gamma_0$  has the dimension of a rate. (If one prefers an explicitly normalized kernel, define  $\bar{K}_\xi := K_\xi/V_\xi$  so that  $\int d^4 z \bar{K}_\xi(z) = 1$ .) This choice yields a positive, integrable weight and, when composed with Hermitian jump operators, preserves complete positivity of the dissipative channel.

**A.4 WESH–Noether normalization.** Coarse-grained consistency imposes a matching condition on the intensity scale:

$$\nu \simeq \gamma_0 \bar{C}_x, \quad \bar{C}_x := \frac{1}{V_\xi} \int d^4 y K_\xi(x - y) C(\Psi; x, y), \quad V_\xi = \int d^4 z K_\xi(z) \sim \xi^4. \quad (\text{A.5})$$

Here  $C(\Psi; x, y)$  is the state-dependent bilocal factor of the WESH channel and  $V_\xi$  is the effective correlation 4-volume. With the normalized measures of Eq. (1.14),  $V_\xi$  is tracked consistently through the coarse-graining;  $\gamma_0$  remains the fundamental

rate scale, while  $\nu$  is fixed by self-consistent matching (hence not independently tunable).

**A.5 Planck anchoring.** With  $m_T \sim M_P$  one has  $\xi \sim L_P$  by (A.3). Combining this with the matching  $\nu \simeq \gamma_0 \bar{C}_x$  (Eq. (A.5)), with  $\bar{C}_x = \mathcal{O}(1)$  on correlated domains, the natural anchoring is

$$\gamma_0 = \Theta_0 t_P^{-1}, \quad \Theta_0 = \mathcal{O}(1). \quad (\text{A.6})$$

Here  $\Theta_0$  collects normalized cumulants and geometric factors from the time sector, ensuring that the anchoring introduces no additional free parameters or arbitrary scales.

**Remarks.** (i) The limit  $\xi \rightarrow 0$  recovers a local channel; finite  $\xi$  implements coarse-grained finite-range locality with finite correlation volume. (ii) The static Yukawa form is a spatial illustration only; the operative kernel is the 4D  $L^1$ -integrable  $K_\xi$  with finite-range support. (iii) The construction is minimal and sufficient to support the collapse kernel of Eq. (1.20) and the no-signaling statement Eq. (1.24), while fixing  $\gamma_0$  via Eq. (A.5).

## Appendix B: Parameter $s$ , Emergence of Physical Time, and the Conceptual Limits of a Language Within Time

### B.1 The Auxiliary Parameter $s$ and its Role in QFTT–WESH

In QFTT–WESH, an intrinsic parameter  $s$  is introduced to write the dynamics of a closed, timeless quantum universe. The universal state  $\rho(s)$  is evolved in  $s$  but  $s$  is *not* an observable: it is a non-physical ordering label used to formulate the GKSL evolution and to state constraints in the WDW sector. “Gauge-like” here means *non-observable scaffolding*; No physical observable depends on the choice of the  $s$ -parametrization: one may view  $s$  as a gauge-like ordering label (any monotone reparametrization is absorbed by the map  $t(s)$ ). This addresses the *problem of time* in a Wheeler–DeWitt universe.

The  $s$ -evolution uses the WESH generator referenced in Eq. (1.15), with  $\hat{H}_{\text{eff}}$ ,  $\mathcal{D}[\cdot]$ ,  $L_{xy}$  and  $\gamma$  given in Eqs. (1.17)–(1.20).

*Notation.* We adopt the operator and kernel conventions of Eqs. (1.17)–(1.20); below we recall only the finite-range support needed for the rates.

**Finite-range kernel.** We keep the *exponential finite-range* dissipative weight (see Eq. (1.20))

$$\gamma(x, y) = \frac{\gamma_0}{N^2} e^{-d(x, y)/\xi} \mathbb{K}_{N_\xi}(x, y).$$

Here  $d(x, y)$  denotes the invariant separation on the label-space; in the IR regime where the emergent metric is available, this coincides with the causal separation (e.g. via Synge’s world function  $\sigma(x, y)$  once  $g_{\mu\nu}^{(T)}$  is defined; see Remark 1.3).

We reserve ‘Yukawa’ for the *mediator* kernel

$$K(x, y) \propto \frac{e^{-|x-y|/\xi}}{4\pi|x-y|} \mathbb{K}_{N_\xi}(x, y).$$

This choice (Eq. (1.20)) avoids a short-distance  $1/r$  singularity in the GKSL rates, keeps an  $L^1$ -integrable weight with finite correlation volume  $\sim \xi^4$ , and, via  $\mathbb{K}_{N_\xi}$ , yields emergent no-signaling in the geometric regime (Eq. (1.24)).

## B.2 Eigentimes and the Emergence of Physical Time $t$

Eigntime events are the spontaneous collapses that make the arrow of time intrinsic. Physical time  $t$  is the coarse-grained monotone of  $s$ :

$$\frac{dt}{ds} = \Gamma[\Psi(s)], \quad t(s) = \int_0^s \Gamma[\Psi(s')] ds'$$

The time-production functional is the dimensionless quantity defined in Eq. (1.33) using the normalized measures of Eq. (1.14):

$$\Gamma[\Psi] = \tau_{\text{Eig}} \left[ \nu \int_x \text{Tr}[\tilde{T}^2(x) \rho \tilde{T}^2(x)] + \int_{xy} \gamma(x, y) C(\Psi; x, y) \text{Tr}[L_{xy}^\dagger L_{xy} \rho] \right],$$

so that  $dt/ds = \Gamma[\rho] \geq 0$ , with  $\Gamma > 0$  under nondegeneracy (Lemma 1.1), and the local/bilocal matching is set by the coarse-grained consistency condition (Eq. (A.5)). Strict positivity of  $\Gamma$  (monotonicity  $dt/ds > 0$ ) follows under Lemma 1.1; eigentime activation on chronogenetic segments under Lemma 1.2.

### Granularity $\rightarrow$ continuum (LLN).

The law of large numbers links event counting to physical time:

$$t(s) \approx \tau_{\text{Eig}} N_{\text{Eig}}(s) \quad (\text{LLN}) \quad (\text{B.1})$$

More precisely:

$$\begin{aligned} N_{\text{Eig}}(s) &= \int_0^s \frac{\Gamma[\Psi(u)]}{\tau_{\text{Eig}}} du + M(s), \\ t(s) &= \int_0^s \Gamma[\Psi(u)] du, \quad \frac{t(s)}{\tau_{\text{Eig}} N_{\text{Eig}}(s)} \xrightarrow{\mathbb{P}} 1. \end{aligned}$$

with martingale  $M(s)$  and  $\tau_{\text{Eig}} \equiv 1/\nu$ . Hence, when  $\mathbb{E}[N_{\text{Eig}}] \gg 1$ , discrete eigentimes smooth to classical  $t$  by the law of large numbers.

**Operator-ordering vs. moments (reader guardrail).** The intensity functional  $\Gamma[\Psi]$  contains the GKSL trace form  $\text{Tr}[\hat{T}^2 \rho \hat{T}^2]$ , *not* the raw moment  $\langle \hat{T}^4 \rangle$ ; the two coincide only if  $[\rho, \hat{T}^2] = 0$ . The trace form is required by complete positivity and appears in the definition of  $\Gamma[\Psi]$  (Eq. (1.33)).

**Example (translation rule  $s \rightarrow t$ ).** A temporalized sentence like “collapse occurs *before* the measurement” means: there exist  $s_{\text{col}} < s_{\text{meas}}$ ; mapping by (1.32) gives  $t_{\text{col}} < t_{\text{meas}}$ . “Before/after” acquires meaning only *after* this mapping.

## B.3 Comparison to Other Frameworks and Interpretations

Having established the QFTT–WESH mechanism for time emergence, it is natural to ask how this approach relates to other frameworks addressing the problem of

time in quantum gravity. Since this appendix defines the auxiliary parameter  $s$ , the eigentime mechanism, and the  $s \rightarrow t$  bootstrap, it is the appropriate place to clarify what each approach *assumes* versus what it *derives*.

Existing frameworks can be grouped by their treatment of time:

- (A) *Time as external parameter.* Standard quantum mechanics, GRW (1986), CSL (Pearle, 1989), and Penrose OR (1996) treat  $t$  as an assumed external time parameter (not derived from a timeless constraint). Collapse models introduce stochastic, time-asymmetric dynamics but do not address black-hole entropy.
- (B) *Time as relational/frozen.* Page & Wootters (1983) address the Wheeler–DeWitt frozen-time problem by treating time as a conditional correlation within a globally static state. The approach is unitary and does not produce an intrinsic arrow of time or dynamical gravity.
- (B') *Time as thermal flow.* The Connes–Rovelli thermal time hypothesis (TTH, 1994) identifies physical time with the modular flow of a KMS state on a von Neumann algebra. The proposal is conceptually elegant but requires a preferred thermal state as input; in QFTT–WESH, the KMS structure is not assumed but emerges at the fixed point of the dissipative dynamics, in the near-horizon regime.
- (C) *Emergent gravity from thermodynamic or holographic postulates.* Jacobson (1995, 2016), Verlinde (2011), and holographic approaches (AdS/CFT, RT, ER=EPR) relate or encode aspects of gravitational dynamics and geometry to entropy or entanglement considerations, assuming either an area/entropy postulate or a microscopic notion of time as input.

Table 5 compares these approaches along four axes: treatment of time, collapse mechanism, arrow of time, and horizon thermodynamics.

| Framework                 | Treatment of Time                       | Collapse Mechanism                               | Arrow of Time     | $S_{\text{BH}}$ Status           |
|---------------------------|-----------------------------------------|--------------------------------------------------|-------------------|----------------------------------|
| Page–Wootters (1983)      | Relational (timeless global constraint) | None (unitary)                                   | Not intrinsic     | Not addressed                    |
| Connes–Rovelli TTH (1994) | Modular flow of KMS state               | None (algebraic)                                 | Thermal (assumed) | Not addressed                    |
| GRW (1986); CSL (1989)    | External parameter $t$                  | Stochastic collapse (phenomenological constants) | Built-in          | Not addressed                    |
| Penrose OR (1996)         | Assumed parameter (not WDW-derived)     | Gravity-related objective reduction              | Built-in          | Not addressed                    |
| Jacobson (1995, 2016)     | Semiclassical (local horizons/balls)    | None (equilibrium condition)                     | Thermodynamic     | <b>Input</b> (area/entanglement) |
| Verlinde (2011)           | Assumes microscopic time                | None (entropic-force picture)                    | Thermodynamic     | <b>Input</b> (holographic)       |
| AdS/CFT, ER=EPR           | Boundary QFT time                       | None (unitary boundary)                          | State-dependent   | Computed holographically         |
| <b>QFTT–WESH</b>          | <b>Emergent</b> via eigentimes          | <b>State-dependent</b> $\Gamma[\Psi]$            | <b>Intrinsic</b>  | <b>Derived</b> (future work)     |

**Table 5:** Comparison of selected approaches to the problem of time and black-hole entropy. Among the frameworks listed, QFTT–WESH is the only one that starts from a timeless Wheeler–DeWitt constraint and derives both the time-emergence mechanism and the structure of the dissipative generator within a single chain, without postulating a microscopic time as an independent input. The infrared consequences (Einstein dynamics and horizon thermodynamics) lie beyond the present scope.

With this landscape clarified, Sec. B.4 summarizes the technical status of the  $s \rightarrow t$  construction.

## B.4 Concluding Remarks

*Finite- $N$  imprint.* Residual discreteness at finite  $N$  should leave a measurable imprint in the intensity channel: the variance and cross-correlation structure of  $\Gamma$  (Eqs. (1.29), (1.33), (1.34)) would provide a direct handle on the granularity of the eigentime mesh and on the interaction range  $\xi$ .

*Bridge to gravity.* At the stationary fixed point selected by the WESH monotone, the gradient-alignment condition  $\partial_\mu \tau = k \partial_\mu \Phi$  holds. Combined with hidden-sector cancellation, this yields the emergent Einstein equation with matter. The full derivation lies beyond the present scope.

*Pre-geometric consistency (App. D).* The foundational constraint is WESH–Noether path independence, not thermodynamic detailed balance; the latter emerges only in the near-horizon KMS regime, after physical time has crystallized.

## B.5 Conceptual Limits and the Paradox of Inherited Language

Inevitably, a theory like QFTT–WESH, aiming to position itself upstream of the 4 dimensions, collides with the epistemological problem of using a language within the temporal line. Concepts such as ‘before’, ‘after’, ‘cause’, and ‘effect’ (but even more implicit assumptions and morphemes) are foundational to our linguistic and logical structures; yet the theory must describe a reality that exists at a more fundamental level than their emergence. At the same time, we identify a taxonomic challenge, implicitly calling for the development of a new mathematical language, perhaps more circular, topological, or inherently atemporal. In the present work, such tension forces a deliberate methodological strategy: to remain communicable, the theory must adopt a temporalized narrative for its exposition, even as its formalism in part describes an atemporal process. The auxiliary parameter  $s$  is the controlled semantic scaffold, a *necessary conceptual unit* that dissolves once the formalism is established, essential to navigate this paradox. We could see it in such a way that any temporal sentence admits an  $s$ -label version (ordering only), then is translated to a  $t$ -statement via Eq. (1.32); Appendix B’s LLN line justifies the continuous limit for macroscopic descriptions. This keeps the exposition communicable while ensuring that predictions depend solely on the emergent, measurable  $t$ . Here lies the epistemological challenge, but at the same time the beauty of this theory.

It is precisely this tension that has motivated a parallel line of investigation, already underway, into what we call Quantum Self-Interacting Poisson Structures

(QSIPS). The guiding intuition is that the state-dependent dissipative dynamics at the heart of WESH may not be a patched modification of existing frameworks but rather the expression of a more primordial mathematical structure, one in which feedback-dependent intensity processes are not anomalies requiring linearization but constitutive features of the process class itself. Whether such a structure can absorb the linguistic paradox identified above, by providing a formalism that is inherently atemporal at its foundations, remains an open and fascinating question.

## Appendix C: Heuristic derivations for $N$ -scaling

**Setup.** The WESH bilocal channel uses a finite-range, normalized kernel with prefactor  $N^{-2}$  (Eq. (1.20)); the intensity channel is quadratic in  $\hat{T}^2$  (Eq. (1.33)). The mediator kernel entering  $\Phi$  is Yukawa (Appendix A), with range  $\xi$  and correlation time  $\tau_{\text{corr}} \sim \xi/c$  (independent of  $N$ ).

**H1 — Pairwise normalization.** Let the effective pair couplings be  $\gamma_{ij} = \gamma_0 N^{-2} w_{ij}$  with  $w_{ij} = \mathcal{O}(1)$  and  $\sum_{i < j} w_{ij} = \Theta(N^2)$  (bounded, finite-range weights; equivalently  $\langle w \rangle = \mathcal{O}(1)$  over  $\Theta(N^2)$  pairs). Then

$$\sum_{j \neq i} \gamma_{ij} = \mathcal{O}(\gamma_0/N), \quad \sum_{i < j} \gamma_{ij} = \mathcal{O}(\gamma_0).$$

Interpretation: the per-site collapse power scales as  $1/N$  while the total power remains  $\mathcal{O}(1)$ , consistent with the  $N^{-2}$  prefactor in Eq. (1.20).

**H2 — Local hazards from block averaging.** For block averages  $X_B = \frac{1}{|B|} \sum_{i \in B} X_i$  (with  $X_i$  a bounded local contribution to the  $\hat{T}^2$ -intensity) and short-range correlations (range  $\xi$ ),

$$\text{Var}(X_B) = \mathcal{O}(|B|^{-1}).$$

Under WESH weighting, the relevant block tracks the entangled set size,  $|B| \sim N$ . Because the intensity channel is quadratic in  $\hat{T}^2$  (Eq. (1.33)), the coarse-grained hazard inherits a square on the variance,

$$\lambda \propto \text{Var}(X_B)^2 = \mathcal{O}(N^{-2}).$$

Thus  $\lambda = \mathcal{O}(N^{-2})$  implies  $\tau_{\text{coh}} \sim \lambda^{-1} \propto N^2$ , while the per-site rate  $\Gamma_i \sim \gamma_0/N$  aligns with H1.

**H3 — Correlation time vs. eigentime spacing.** The kernel range fixes  $\tau_{\text{corr}} \sim \xi/c$  (independent of  $N$ ), while the effective macroscopic inter-event spacing grows



under H1–H2 as  $\tau_{\text{Eig}}^{(\text{eff})} \propto N^2$  (see Lemma 1.3). Hence

$$\mu := \frac{\tau_{\text{corr}}}{\tau_{\text{Eig}}^{(\text{eff})}} = \mathcal{O}(N^{-2}),$$

placing the dynamics deep in the Markov window for large  $N$ .

**Consequence.** H1–H3 jointly underwrite the collective–stability law  $\tau_{\text{coh}}(N) \propto N^2$  and the small–parameter regime  $\mu = \mathcal{O}(N^{-2})$  used in the variational analysis (Lemma 1.3).

## Appendix D — WESH–Noether as Pre-Geometric Path Independence

**Context.** The Wheeler–DeWitt sector carries no physical time, metric, or thermodynamics. The foundational constraint developed here is a *pre-geometric* conservation law compatible with the WESH generator ((1.15)). All references to “flow” denote ordinal evolution in state space; no temporal structure is assumed prior to the bootstrap.

**Remark D.1** (Static vs dynamic in the pre-geometric regime). *In the WDW sector, states are ordered by the Lyapunov functional  $\mathcal{M}[\rho]$ , global charges  $\langle \hat{Q}_{\text{tot}} \rangle$  remain constant along any WESH flowline, and the gradient  $\nabla_{\rho} \mathcal{M}$  defines potential streamlines. Physical time emerges when this configuration is actualized via the bootstrap  $dt/ds = \Gamma[\Psi] > 0$ , converting potential order into temporal flow.*

### D.1 Operator formulation of pre-geometric Noether

Let  $\mathcal{L}$  be the WESH GKSL generator (see (1.15)) and  $\mathcal{L}^{\dagger}$  its adjoint on observables:

$$\mathcal{L}^{\dagger}[A] = -i[H_{\text{eff}}, A] + \nu \int_x \mathcal{D}_{\hat{T}^2(x)}^{\dagger}[A] + \int_{xy} \gamma(x, y) C(\Psi; x, y) \mathcal{D}_{L_{xy}}^{\dagger}[A],$$

with  $\int_x, \int_{xy}$  as in Eq. (1.14),  $\mathcal{D}_L^\dagger[A] = L^\dagger AL - \frac{1}{2}\{L^\dagger L, A\}$ , and  $L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)$  (Hermitian).

**Definition D.1 (Pre-geometric WESH–Noether).** For every global conserved charge  $\hat{Q}_{\text{tot}}$ ,

$$\mathcal{L}^\dagger[\hat{Q}_{\text{tot}}] = 0. \quad (\text{D.1})$$

This is equivalent to  $\frac{d}{ds}\langle\hat{Q}_{\text{tot}}\rangle = 0$  for all  $\rho(s)$ . The kernel weight is exponential with finite range  $\xi$  (see (1.20)).

**Remark D.2 (Self-consistent physical subset).** The “physical subset” is not pre-imposed. Operationally, it is the basin of attraction of the bootstrap fixed point(s)  $\rho^*$  that (i) minimize the Lyapunov functional (Lemma 1.3), (ii) satisfy WESH–Noether at the fixed point, and (iii) are reached by iterating the bootstrap from RAQ-physical initial conditions (i.e., they belong to the basin of attraction). WESH–Noether thus acts as a local structural constraint of the (state-dependent) generator throughout the evolution, without a pre/post-bootstrap dichotomy.

**Theorem D.1 (WESH–Noether).** Let  $\mathcal{L}$  be a GKSL generator governing evolution in the atemporal ordering parameter  $s$  within the Wheeler–DeWitt sector, with Heisenberg adjoint  $\mathcal{L}^\dagger$ . Assume Hermitian jump operators  $L_\alpha \in \{\hat{T}^2(x), L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)\}$  and the mild spectral regularity of Proposition D.2. Then, for any global charge operator  $\hat{Q}_a$ , the following are equivalent:

$$\mathcal{L}^\dagger[\hat{Q}_a] = 0 \quad \Longleftrightarrow \quad [H_{\text{eff}}, \hat{Q}_a] = 0 \text{ and } [L_\alpha, \hat{Q}_a] = 0 \text{ for all } \alpha \text{ with nonzero rate.}$$

No geometric or thermal structure is presupposed.

**Corollary D.1 (Path independence /  $s$ -conservation).** For all states  $\rho(s)$  and all  $a$ ,  $\frac{d}{ds}\langle\hat{Q}_a\rangle = 0$ . Equivalently, the charge one-form  $\omega_a[\rho] := \text{Tr}(\hat{Q}_a \mathcal{L}[\rho]) ds$  is exact (zero holonomy).

**Remark D.3 (Specialization to the KMS wedge).** In the near-horizon Rindler wedge, Modular–KMS spectral regularity implies the hypothesis of Prop. D.2; hence the equivalence holds there verbatim.

## D.2 Necessary & sufficient algebraic conditions (Hermitian Lindblad structure)

In the present framework the Lindblad operators are Hermitian,  $L \in \{\hat{T}^2(x), L_{xy}\}$ , and the standard GKSL identity

$$\mathrm{Tr}(A \mathcal{D}[L]\rho) = \frac{1}{2} \langle L^\dagger[A, L] + [L^\dagger, A]L \rangle$$

yields, for Hermitian  $L$ ,

$$\mathcal{D}_L^\dagger[\hat{Q}] = -\frac{1}{2} [L, [\hat{Q}, L]]. \quad (\text{D.2})$$

### Proposition D.2 (Equivalence under mild spectral regularity).

Assume:

- (i)  $\hat{Q}_{\text{tot}}$  is bounded (or the identities are understood as quadratic forms on a common invariant core);
- (ii) Hermitian jump set  $L \in \{\hat{T}^2(x), L_{xy}\}$  in the generator (1.15);
- (iii) (*Mild spectral regularity*) For each such  $L$ , the global charge  $\hat{Q}_{\text{tot}}$  is block-diagonal in the spectral decomposition of  $L$  (equivalently:  $\hat{Q}_{\text{tot}}$  commutes with the spectral projectors of  $L$  on the relevant support).

Then the pre-geometric WESH-Noether condition (D.1) holds *iff*

$$[H_{\text{eff}}, \hat{Q}_{\text{tot}}] = 0, \quad [\hat{T}^2(x), \hat{Q}_{\text{tot}}] = 0 \quad \forall x, \quad [L_{xy}, \hat{Q}_{\text{tot}}] = 0 \quad \forall x, y. \quad (\text{D.3})$$

*Proof. Sufficiency.* If (D.3) holds, each dissipative term in  $\mathcal{L}^\dagger[\hat{Q}_{\text{tot}}]$  vanishes by (D.2), and the Hamiltonian part yields  $[H_{\text{eff}}, \hat{Q}_{\text{tot}}] = 0$ , hence (D.1).

*Necessity.* Assume  $\mathcal{L}^\dagger[\hat{Q}_{\text{tot}}] = 0$  with  $\hat{Q}_{\text{tot}} = \hat{Q}_{\text{tot}}^\dagger$ . Take the Hilbert-Schmidt pairing with  $\hat{Q}_{\text{tot}}$ :

$$0 = \mathrm{Tr}(\hat{Q}_{\text{tot}} \mathcal{L}^\dagger[\hat{Q}_{\text{tot}}]).$$

The Hamiltonian contribution vanishes by cyclicity,  $\mathrm{Tr}(\hat{Q}_{\text{tot}}[-iH_{\text{eff}}, \hat{Q}_{\text{tot}}]) = 0$ . For Hermitian Lindblad operators, one has

$$\mathrm{Tr}(\hat{Q}_{\text{tot}} \mathcal{D}_L^\dagger[\hat{Q}_{\text{tot}}]) = -\frac{1}{2} \mathrm{Tr}([L, \hat{Q}_{\text{tot}}]^\dagger [L, \hat{Q}_{\text{tot}}]) \leq 0.$$

Using  $\nu > 0$  and  $\gamma(x, y) C(\Psi; x, y) \geq 0$ , this yields

$$\int_x \text{Tr}([\hat{T}^2(x), \hat{Q}_{\text{tot}}]^\dagger [\hat{T}^2(x), \hat{Q}_{\text{tot}}]) = 0,$$

$$\int_{xy} \gamma(x, y) C(\Psi; x, y) \text{Tr}([L_{xy}, \hat{Q}_{\text{tot}}]^\dagger [L_{xy}, \hat{Q}_{\text{tot}}]) = 0,$$

hence  $[\hat{T}^2(x), \hat{Q}_{\text{tot}}] = 0$  (on the support of  $\int_x$ ) and  $[L_{xy}, \hat{Q}_{\text{tot}}] = 0$  wherever  $\gamma(x, y) C(\Psi; x, y) \neq 0$ . Plugging back into  $\mathcal{L}^\dagger[\hat{Q}_{\text{tot}}] = 0$  leaves  $-i[H_{\text{eff}}, \hat{Q}_{\text{tot}}] = 0$ , hence  $[H_{\text{eff}}, \hat{Q}_{\text{tot}}] = 0$ .  $\square$

**Remark D.4** (Athermal generality). *Proposition D.2 is pre-geometric and athermal: no KMS structure, thermal equilibrium, or wedge geometry is required. The proposition relies solely on mild spectral regularity of the Lindblad operators in the kinematical Hilbert space. The near-horizon Rindler wedge provides a concrete regime where the hypothesis of Proposition D.2 is satisfied; the wedge analysis is thus a specialization of the general algebraic framework established here, not an alternative foundation.*

**Lemma D.1** (Spectral genericity). *For the time-field operator  $\hat{T}(x)$  with continuous spectrum in the WDW sector, the quadratic operator  $\hat{T}^2(x)$  has a non-degenerate spectral measure on any bounded interval  $[E_{\min}, E_{\max}]$  with probability 1 under generic local perturbations.*

*Proof sketch.* The spectrum of  $\hat{T}^2(x)$  is the image of  $\sigma(\hat{T})$  under  $\lambda \mapsto \lambda^2$ . For continuous  $\sigma(\hat{T}) \subset \mathbb{R}$ , generic perturbations preserve measure-theoretic nondegeneracy (Kato perturbation theory), which supports the projector-commutation regularity used in Prop. D.2.  $\square$

**Remark D.5** (Regularity note). *Commutation with spectral projectors is the minimal operational hypothesis required by the double-commutator identity; it is generic for local polynomials of  $\hat{T}$  and stable under local perturbations. Since  $L_{xy} = \hat{T}^2(x) - \hat{T}^2(y)$ ,  $[L_{xy}, \hat{Q}_{\text{tot}}] = 0$  follows by linearity once  $[\hat{T}^2(x), \hat{Q}_{\text{tot}}] = [\hat{T}^2(y), \hat{Q}_{\text{tot}}] = 0$ .*

*The mild spectral regularity is generic for local polynomials of  $\hat{T}$  and their differences. It is strictly weaker than assuming any thermal structure and does not presuppose the emergence of  $t$ .*

**Remark D.6** (T-neutrality as a consistency requirement). *If  $[\hat{T}, \hat{Q}_{\text{tot}}] \neq 0$ , the state-dependent coefficients of the generator induce a charge-dependent intensity; the ordinal speed  $dt/ds$  becomes background-dependent, spoiling the universality of physical time. Moreover, along different state-space paths one generically sources  $\hat{Q}_{\text{tot}}$ , violating path independence. Hence T-neutrality is not an external postulate but the unique choice compatible with path independence, universality of emergent time, and no-signaling.*

**Remark D.7** (Operator-level vs expectation-level conservation). *The WESH-Noether condition (D.1) is an operator-level statement, not merely conservation of expectations. Operator-level conservation guarantees zero holonomy of the charge one-form on state space; conserving only  $\langle \hat{Q}_{\text{tot}} \rangle$  allows hidden anomalies in the commutant that re-emerge under block decompositions or sub-sector measurements, obstructing the emergence of a unique geometry.*

### D.3 From potential order to geometry (bridge to alignment)

Let  $\mathcal{M}[\rho]$  be the WESH Lyapunov functional (strictly convex in the regime of interest; see Lemma 1.3). Consider the constrained minimization “minimize  $\mathcal{M}$  subject to WESH-Noether”. At a stationary point, the classical Karush–Kuhn–Tucker (KKT) optimality conditions (Karush, 1939; Kuhn & Tucker, 1951) require that the state-space gradient  $\nabla_{\rho} \mathcal{M}$  lies in the span of the constraint gradients. The KKT framework applies here because the Lyapunov functional is smooth and the WESH-Noether constraints define a regular manifold in state space; the Lagrange multipliers encode the relative weight of each conservation law at the fixed point. For the quadratic structure of  $\mathcal{M}$  (in  $\|\partial_{\mu} \tau\|^2$ ) and the Yukawa-weighted entanglement potential, this forces the field-level relation  $\partial_{\mu} \tau \parallel \partial_{\mu} \Phi$ , i.e.:

$$\partial_{\mu} \tau = k \partial_{\mu} \Phi, \quad \Phi(x) = \int K(x - y) C(\rho^*; x, y) d^4 y, \quad (\text{D.4})$$

i.e. the pre-geometric consistency crystallizes into an emergent metric relation.

**Remark D.8** (Existence vs uniqueness). *WESH-Noether ensures existence of a stationary state (Lemma 1.3; Schauder–Tychonoff) but not uniqueness. Uniqueness follows from additional physical constraints:*

- (i) *Finite-range support of  $\gamma(x, y)$  (emergent no-signaling at the fixed point);*

- (ii) *Finite correlation time*  $\tau_{\text{corr}} \sim \xi/c$  (Markov window);
- (iii) *Cross-horizon connectivity* (null-plane Markov mixing).

*These are not fine-tunings but consistency requirements for the scaling hypothesis  $\xi \simeq L_P$ . Together with WESH–Noether, they select the unique fixed point where gradient alignment (D.4) holds.*

## D.4 Empirical signatures

The pre-geometric constraint yields distinct empirical targets:

1. **Core: collective stability.** The coherence time scales as  $\tau_{\text{coh}}(N) \propto N^2$ , reflecting  $N_\xi$ -local mixing under the WESH constraint.
2. **Diagnostic: angular dependence.** The decoherence rate obeys  $\bar{\Gamma} [1 + \varepsilon \cos^2 \theta]$  with  $\varepsilon = (a_2/a_0) (G_2/G_0)$ , factorizing state anisotropy and device geometry.
3. **Horizon thermodynamics.** Near-horizon KMS structure is expected to yield a thermodynamic law (area term and logarithmic corrections) without external parameters; this analysis lies beyond the present scope.

Each item probes a different layer of the construction: collective stability and black-hole thermodynamics test core structural aspects of WESH (including the IR bridge to gravity), while the angular dependence provides a diagnostic check of finite-range connectivity and state/geometry factorization.

## D.5 Physical meaning and scope

*Meaning.* Eqs. (D.1)–(D.3) formalize the “no-vorticity” intuition without thermodynamic inputs: in the minimal GKSL representation each Hermitian Lindblad channel commutes with the global charges, hence the WESH flow cannot generate spurious global sources. *Scope.* Detailed balance is *derived* at the emergent level (near-horizon KMS), not assumed.

**Laminar-flow analogy.** Imagine a stationary laminar fluid where the velocity field is globally time-independent yet locally directed along well-defined streamlines: this captures the pre-geometric WESH structure in state space. “No-vorticity” corresponds to path independence of global charges (zero holonomy of  $\omega_Q$ ); stagnation points correspond to eigentimes where geometric structure crystallizes. Physical time  $t$  then emerges as the local flow speed via  $dt/ds = \Gamma[\Psi]$ , actualizing

the potential streamlines into worldlines. This clarifies why classical detailed balance cannot be imposed at this stage: temperature and KMS structure arise only after the flow has crystallized into a geometry.

**RAQ compatibility.** The WESH dynamics preserves the RAQ-physical subspace: T-neutrality and Hermitian jumps ensure that the generator neither sources nor leaks gauge constraints. The global WESH-Noether condition is the observable face of this preservation in the emergent sector.

## Appendix E — CP/TP preservation and no-signaling in WESH

**Setting.** Consider the pre-geometric master equation (1.15) with Hermitian channels

$$L_\alpha \in \{ \hat{T}^2(x), L_{xy} = \hat{T}^2(x) - \hat{T}^2(y) \},$$

and scalar rates

$$c_\alpha(s) \in \{ \nu, \gamma(x, y) C(\Psi; x, y) \}, \quad 0 \leq C < 1, \gamma \geq 0,$$

assumed bounded and piecewise continuous in  $s$  (hence strongly measurable and uniformly bounded on compact intervals). Here  $C(\Psi; x, y)$  depends only on *local/bilocal reductions*

$$\rho_x = \text{Tr}_{x^c} \rho, \quad \rho_y = \text{Tr}_{y^c} \rho, \quad \rho_{xy} = \text{Tr}_{(xy)^c} \rho,$$

through bounded, Borel-measurable functionals that are Lipschitz in the trace norm. The normalized measures  $\int_x$  and  $\int_{xy}$  (Eq. (1.14)) implement a coarse-graining over correlation cells of four-volume  $V_\xi = \mathcal{O}(\xi^4)$ . The effective theory at the cell scale  $\xi \simeq L_P$  operates subject to a UV-cutoff that restricts the physical state space to a sector of bounded local energy density. Within this truncated effective description, the cell-averaged jump operators  $\hat{T}^2(x)$  and  $L_{xy}$  act as bounded operators on the relevant physical domain. The finite-range kernel  $\gamma(x, y)$  has finite  $L^1$  norm  $\|K_\xi\|_{L^1} \sim V_\xi = \mathcal{O}(\xi^4)$ . The GKSL exponentiation in the proof below is therefore well-defined at every level of the construction. The effective Hamiltonian  $H_{\text{eff}}$  is self-adjoint so that  $-i[H_{\text{eff}}, \cdot]$  generates a strongly continuous one-parameter group of trace-norm isometries on the trace class.

**Theorem E.1** (CP/TP preservation for the nonlinear WESH evolution). *Under the above assumptions, for any  $s \geq 0$  the non-autonomous evolution  $E_s : \rho_0 \mapsto \rho(s)$  generated by (1.15) preserves positivity and unit trace; each frozen micro-step is CPTP, and the product-integral limit preserves these properties.*

*Proof sketch.* Partition  $[0, s]$  into  $0 = s_0 < \dots < s_N = s$  and freeze the rates at the left endpoints:

$$\mathcal{L}_k(\cdot) = -i[H_{\text{eff}}, \cdot] + \nu \int d^4x \mathcal{D}[\hat{T}^2(x)](\cdot) + \iint d^4x d^4y \gamma(x, y) C(\Psi_{s_k}; x, y) \mathcal{D}[L_{xy}](\cdot).$$

Each  $\mathcal{L}_k$  is GKSL (Hermitian jumps, nonnegative rates), thus  $\Phi_k := \exp(\Delta s_k \mathcal{L}_k)$  is CPTP and  $E_s^{(N)} = \Phi_{N-1} \circ \dots \circ \Phi_0$  is CPTP by composition.

*Non-autonomous convergence.* Assume  $s \mapsto c_\alpha(s)$  are strongly measurable and uniformly bounded, and that the map  $\rho \mapsto C(\Psi; x, y)$  is Lipschitz in the trace norm via the reduced states. Then the product-integral scheme with frozen coefficients converges strongly to the unique non-autonomous evolution solving (1.15). Strong limits of CPTP maps remain CP and TP, hence  $E_s$  is positivity- and trace-preserving (stepwise CPTP).  $\square$

**Corollary E.1** (System-ancilla consistency). *For any ancilla  $A$  and initial  $\rho_{SA}$ ,  $(E_s \otimes \mathbb{I}_A)(\rho_{SA})$  preserves positivity and unit trace.*

**Proposition E.1** (Emergent no-signaling). *In the emergent geometric regime (where  $N_\xi$ -separation coincides with metric spacelike separation), let  $\Sigma$  be a Cauchy slice and  $A \subset \Sigma$  with complement  $A^c$ . If (i)  $\gamma(x, y) = 0$  outside  $N_\xi$  and (ii)  $[\hat{T}(x), \hat{T}(y)] = 0$  for  $N_\xi$ -separated  $(x, y)$ , then*

$$\frac{d}{ds} \rho_A(s) = \text{Tr}_{A^c} (\mathcal{L}_{N_\xi(A)}[\rho(s)]),$$

*i.e.  $\rho_A$  depends only on operators supported in the  $N_\xi$ -envelope of  $A$ ;  $N_\xi$ -separated operations on  $A^c$  cannot instantaneously affect  $\rho_A$ .*

**Remark E.1** (Nonlinearity vs CP). *Nonlinearity enters only through the scalar rates  $c_\alpha(s)$  (via  $C(\Psi; x, y)$ ), not in the operator set  $\{L_\alpha\}$ . Freezing  $c_\alpha$  yields GKSL steps; product-integral concatenation preserves CP/TP in the limit, avoiding pathologies such as loss of positivity or violation of complete positivity that can occur in arbitrary nonlinear master equations.*



## Appendix F — Consistency selection of the quadratic dissipator $\mathcal{D}[\hat{T}^2]$

**Setting.** Local channels are even, real-analytic functionals of the time field,

$$L_x = F(\hat{T}(x)) = \sum_{m \geq 1} a_{2m} \hat{T}^{2m}(x), \quad L_{xy} = F(\hat{T}(x)) - F(\hat{T}(y)),$$

consistent with CPT-evenness. The operator-level WESH-Noether constraint (App. D, Eq. (D.1)) fixes the commutant structure; here we address the admissible *local* form in the infrared.

### F.1 Constraints and selection principle

**Constraints.**

- (i) **CPT-evenness:** time-reversal on  $\hat{T}$  excludes odd powers.
- (ii) **WESH-Noether:** compatibility with (D.1) throughout the flow.
- (iii) **Collective stability (structural requirement):** the bootstrap closure condition  $\mu \cdot \sigma_\alpha = \mathcal{O}(1)$  (Lemma 1.3) uniquely fixes  $\alpha = 2$ , yielding  $\tau_{\text{coh}}(N) \propto N^2$  (Sec. 2; cf. Eq. (1.12) in Sec. 1). This is a structural consistency requirement of the fixed-regime, not an empirical postulate.

**Proposition F.1** (Quadratic local normal form). *Under (i)–(iii), the admissible local normal form is uniquely quadratic:*

$$F(z) = a_2 z^2, \quad \text{i.e. } \mathcal{D}[\hat{T}^2(x)].$$

*Proof.* CPT symmetry implies that  $F$  is even, hence the leading monomial can be written as  $F(z) \sim z^{2n}$  with  $n \geq 1$ . As shown in Sec. 1 (Eq. (1.12)), under the WESH normalization  $\gamma \propto N^{-2}$  one has

$$F(\hat{T}) \sim \hat{T}^{2n} \quad \Rightarrow \quad \tau_{\text{coh}}(N) \propto N^{2n}.$$

Imposing collective stability  $\tau_{\text{coh}}(N) \propto N^2$  fixes  $n = 1$  uniquely, hence  $F(z) = a_2 z^2$  and the local channel is  $\mathcal{D}[\hat{T}^2(x)]$ .  $\square$

**Remark F.1** (Empirical validation). *The  $N^2$  scaling is used in the theory as a fixed-regime consistency condition (collective stability) selecting the quadratic channel. Experimentally, observing  $\tau_{\text{coh}} \propto N^2$  acts as a direct validation of this structural requirement and excludes higher even monomials in the collective regime.*

**Remark F.2** (Convergence with effective field theory). *The quadratic selection also follows from IR minimality: among CPT-even operators, the lowest-dimension monomial dominates at long wavelengths in the Wilsonian sense. This convergence of two independent arguments, structural (collective stability from bootstrap closure) and effective (Wilsonian IR dominance), strengthens confidence in the uniqueness of  $\mathcal{D}[\hat{T}^2]$ .*

**Remark F.3** (Deeper algebraic foundation). *The constraint-based selection above is complemented by a deeper algebraic argument developed in the main text (Remark 1.1): for  $n \geq 2$ , the bilocal jump operator  $\hat{T}^{2n}(x) - \hat{T}^{2n}(y)$  factorizes as  $[\hat{T}^2(x) - \hat{T}^2(y)] \cdot P_{n-1}$ , and the polynomial prefactor  $P_{n-1}$  contaminates the Lyapunov structure, the field equations, and the hidden-sector cancellation. Only  $n = 1$  yields an irreducible jump. This algebraic rigidity suggests that the quadratic structure may be a manifestation of a more fundamental process class, the Quantum Self-Interacting Poisson Structures (QSIPS), in which state-dependent dissipation rates are not perturbative corrections but constitutive features of the dynamics. A detailed investigation of this broader class is underway.*

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